



A global perspective on social stratification in science

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Working paper



SCAN ME

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Problem



There are [*bibliometric*] myths about academia on **productivity**, **internationalization**, **mobility**, and **impact**.

Problem



There are [*bibliometric*] myths about academia on **productivity**, **internationalization**, **mobility**, and **impact**.

Number of publications has increased a lot. **Collaboration** is the norm (*team science*).

33% of 8 million authors in Scopus have **1** paper. Median N. of authors in 28+ million pubs: **2**, 75% quantile is 4 (**27.2%** solo authored).

High and increased share of **internationalization**

75% of pubs are by authors from one country (national pubs, in OpenAlex: 75%).

Vast and growing number of **mobility** and it positively affects scholars' performance, collaborations and **impact**.

87.5%, one country of affiliation (no international mobility; quite similar in OpenAlex), **73.5%**, one region (no sub-national mobility)

Gap



These processes need to be studied **simultaneously**, in an **integrated** framework and on a **global** scale.

To show that they **might not be as prevalent as the literature states** or the trends differ when the four processes are considered **together**.

Question and Method

Are authors who **collaborate the most**, also the **most mobile, productive, and cited**?

With whom do they collaborate? **Within** their group?

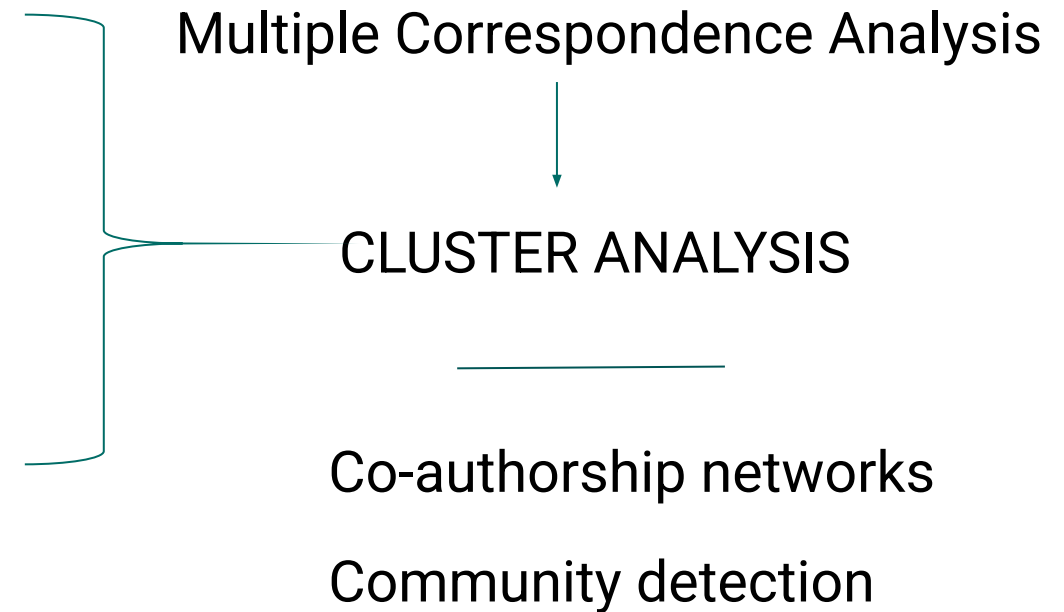
We selected 12 most widely used variables:

4 measuring **collaboration**

3 measuring **mobility**

3 measuring **productivity**

2 measuring **impact**

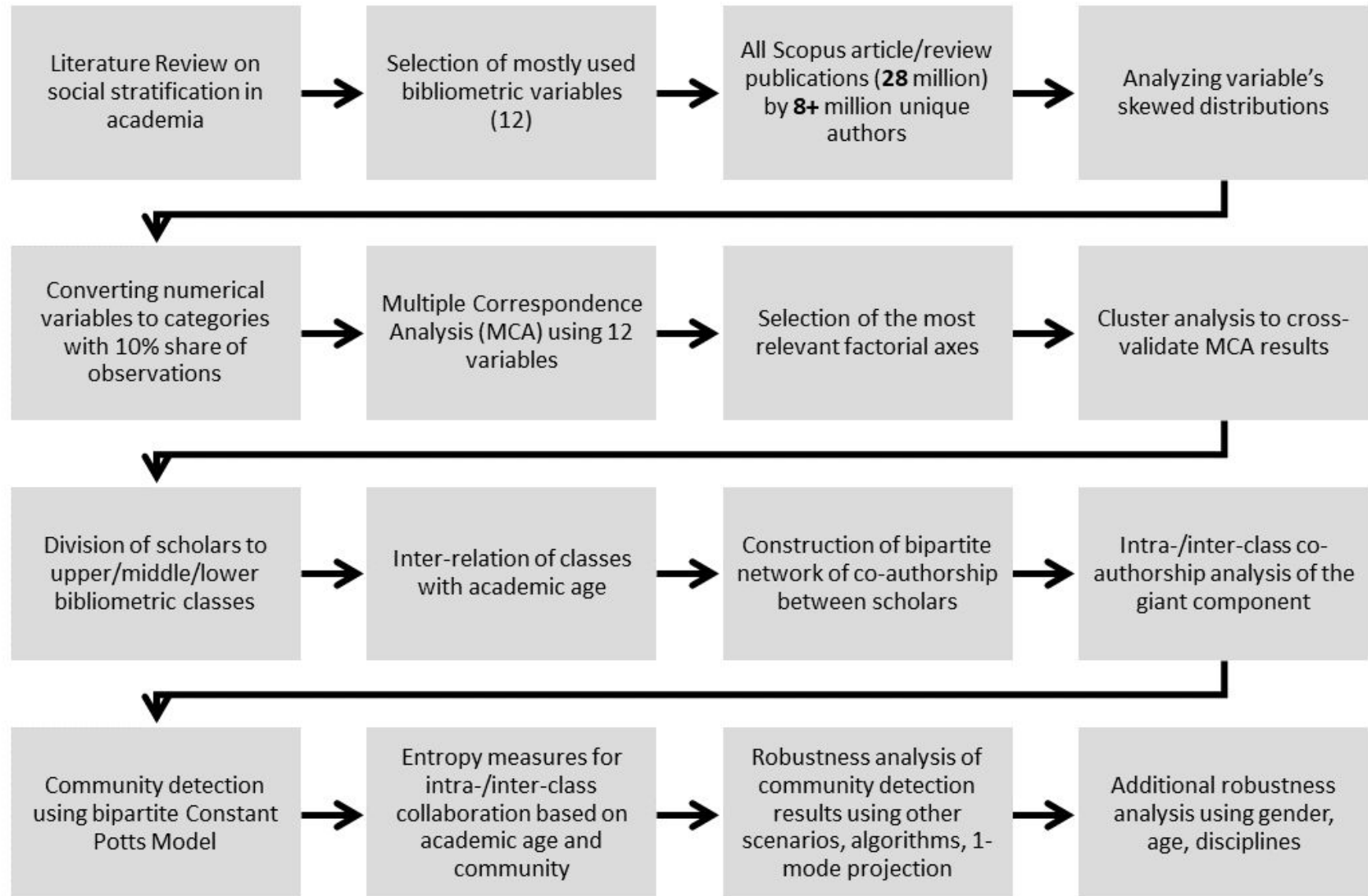


12 mostly used bibliometric variables



Productivity	Number of publications
	Fractional count of publications
	Number of first-author publications
Collaboration/ internationalization	Number of coauthored papers
	Average number of coauthors per paper
	Number of internationally coauthored publications
	Number of nationally coauthored publications
Mobility	Number of international changes in academic affiliation
	Number of national changes in academic affiliation
	Number of affiliated organizations
Impact/visibility	Average number of citations per paper
	Total number of citations

Our data and analysis workflow



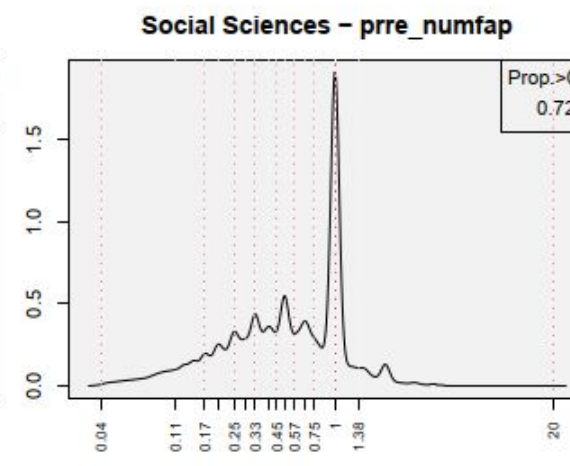
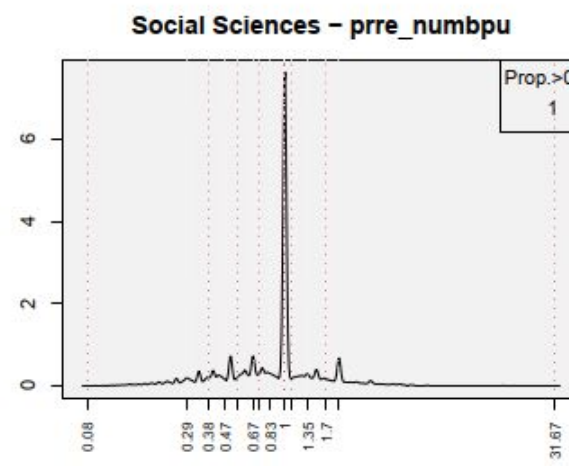
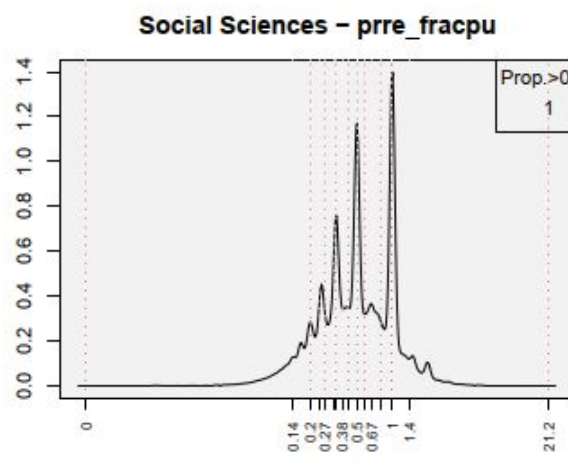
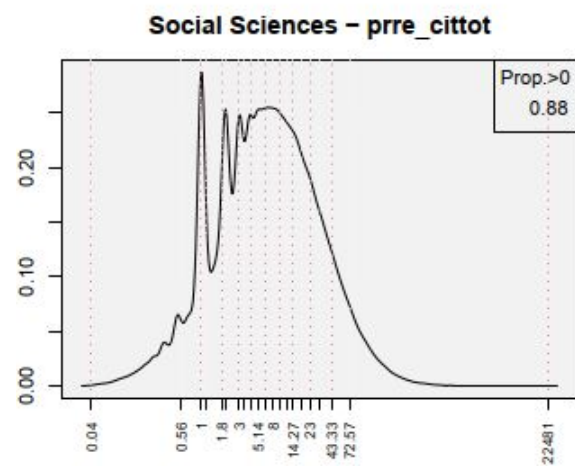
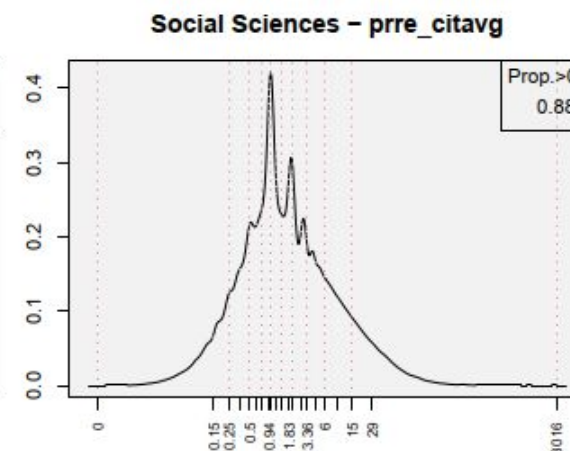
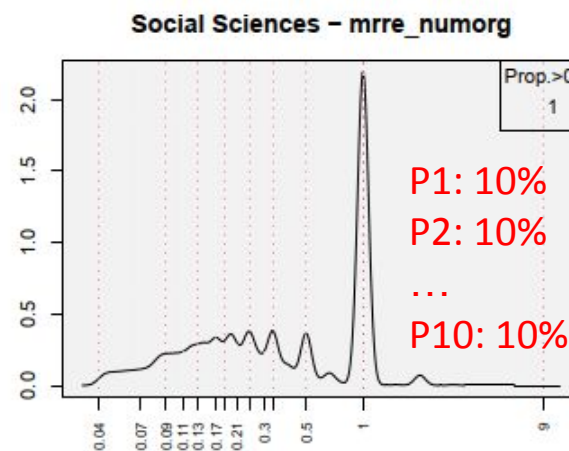
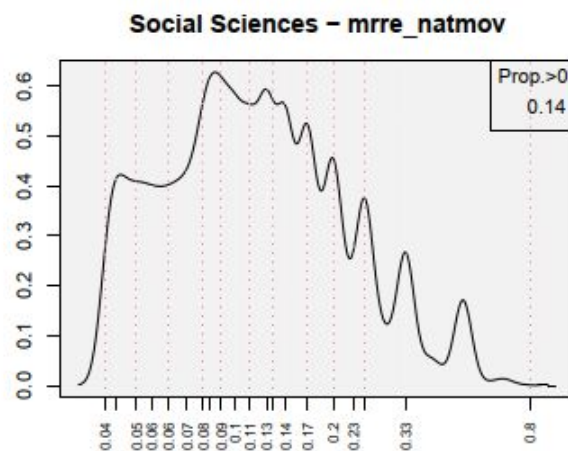
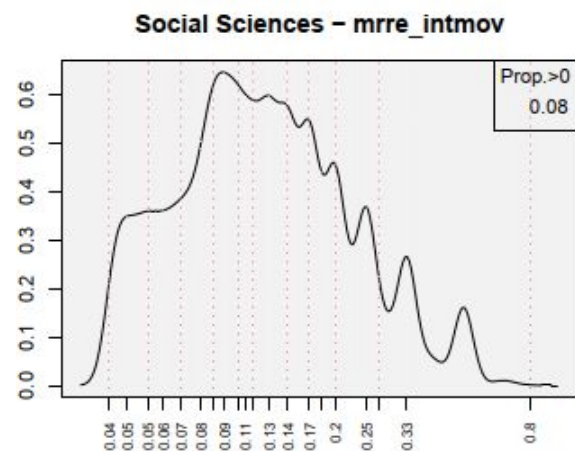
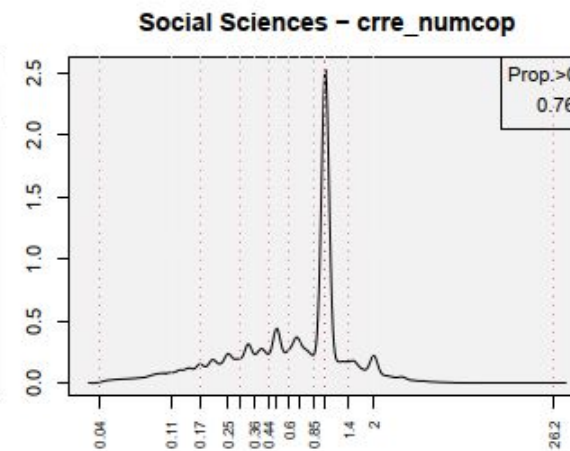
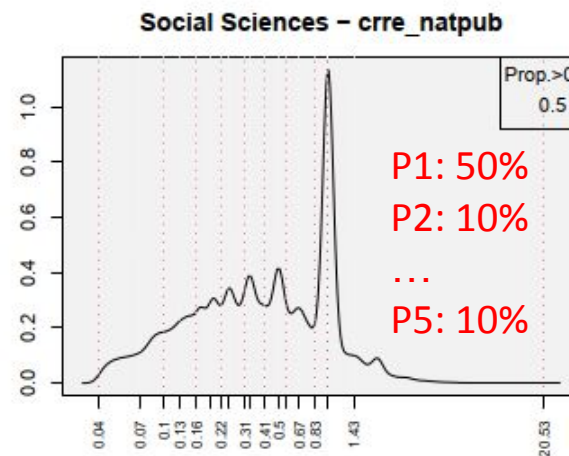
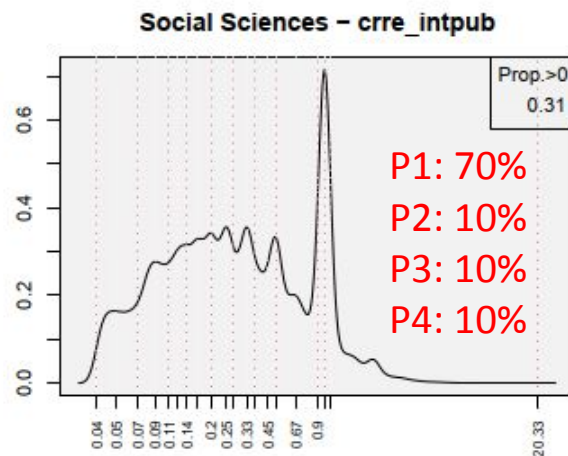
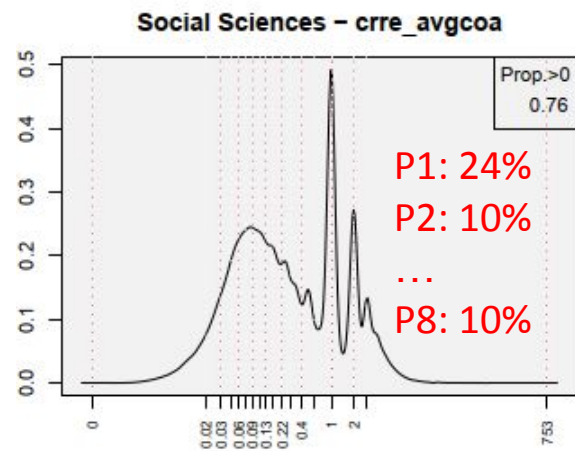
Data



Variable	Scopus**, 1996-2020
N. of publications (article/reviews)	28,461,324
N. of authors (with SCP author ID)	8,225,368
N. of authors (with SCP ID) with 1 publication (33%)	2,709,928
N. of institutions	53,787
N. of authors with 1 institution (64.8%)	5,327,397
N. of countries	205
N. of authors with 1 country (87.5%)	7,200,994
N. of regions	1,836
N. of authors with 1 region (73.5%)	6,042,077
N. of unique years	25
N. of authors with 1 year (36.8%)	3,025,586
N. of papers with 1 author (27.2%)	7,743,400

OECD FOS (+70% pubs, most category)	Count of authors
Agricultural Sciences	237,960
Engineering and Technology	786,541
Humanities	113,054
Medical and Health Sciences	3,122,108
Natural Sciences	3,475,981
No_discipline_assigned	41,278
Social Sciences	448,446

** Data from German Competence Network for Bibliometrics via MPDL





What if we combine these 12 variables to three inter-related axes with MCA?

Contributions

- (i) include **extreme values** in the analysis,
- (ii) capture potential **non-linear relationships** among variables,
- (iii) preserve the **distributional characteristics** of each indicator, and
- (iv) avoids potential biases in correlational analyses due to **outlier observations**.

A **multidimensional measure** of **social stratification** within scientific communities

OECD macro field of science

□ Agricultural Sciences
○ Engineering and Technology

■ Humanities
◆ Medical and Health Sciences

▲ Natural Sciences
◆ Social Sciences

--- Mean contribution

First author publications

Total publications

Fractional pubs.

Total citations

Avg. citations

Num. organizations

Num. nat. moves

Num. intl moves

Num. coauthored pubs.

Num. national pubs.

Num. intl. publications

Avg. collaborations

0 5 10 15 20 25

Percentage contribution to the 1st axis

0 5 10 15 20 25

Percentage contribution to the 2nd axis

0 5 10 15 20 25

Percentage contribution to the 3rd axis

OECD macro field of science

□ Agricultural Sciences

○ Engineering and Technology



■ Humanities

● Medical and Health Sciences

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--- Mean contribution

First author publications

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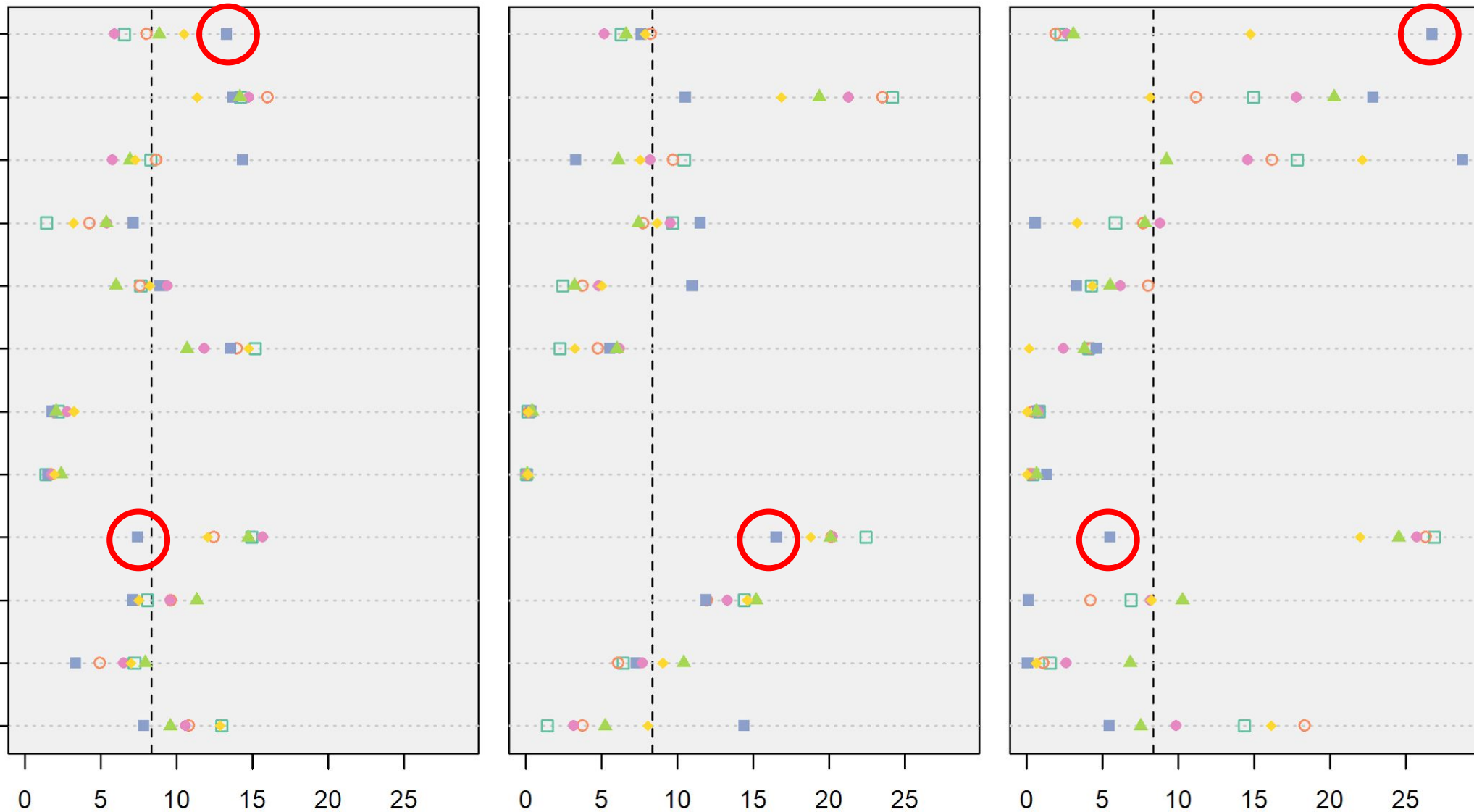
Num. intl. publications

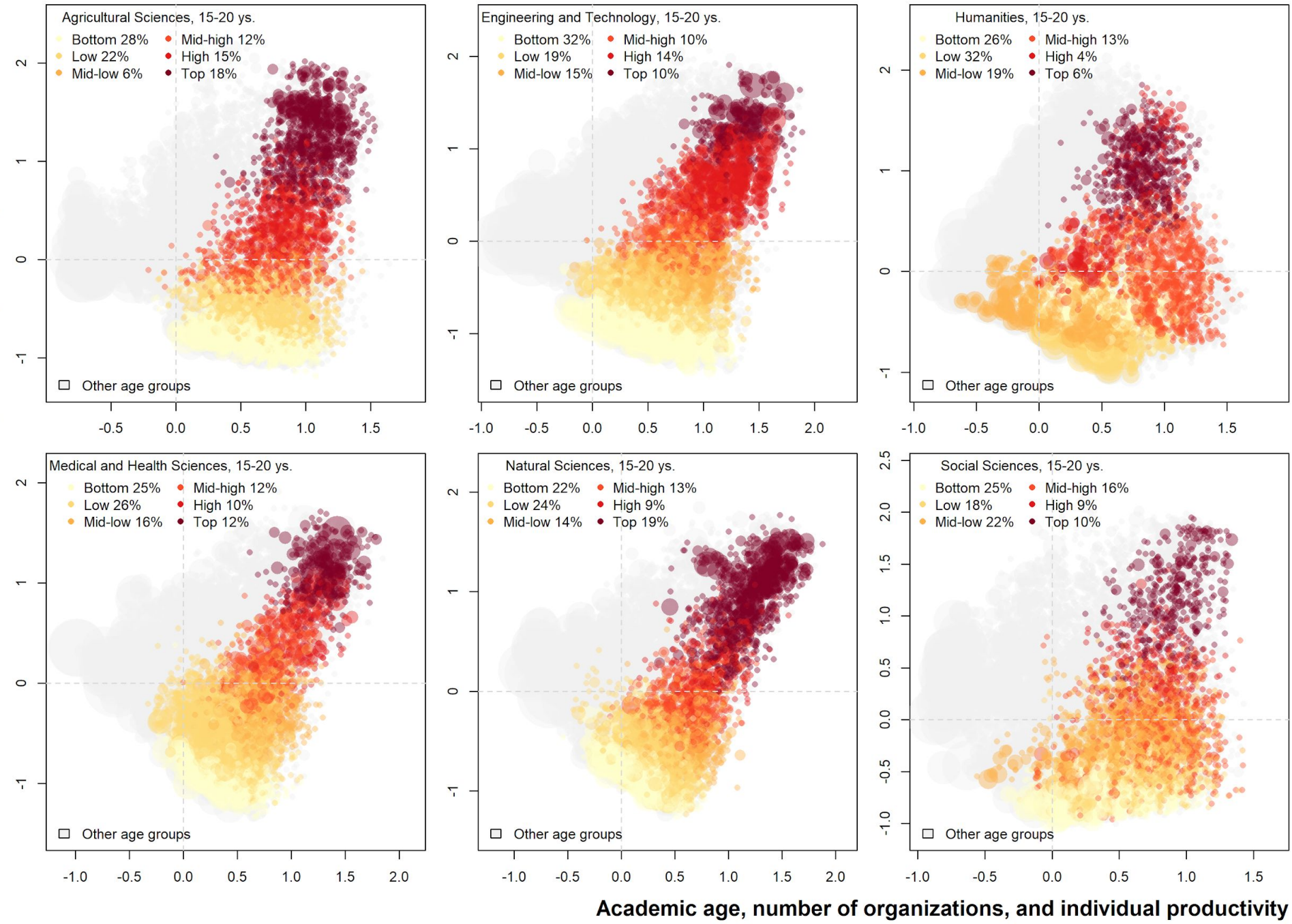
Avg. collaborations

Percentage contribution to the 1st axis

Percentage contribution to the 2nd axis

Percentage contribution to the 3rd axis





15-20 years old

6, 8, 10 clusters were evaluated, 6 was chosen

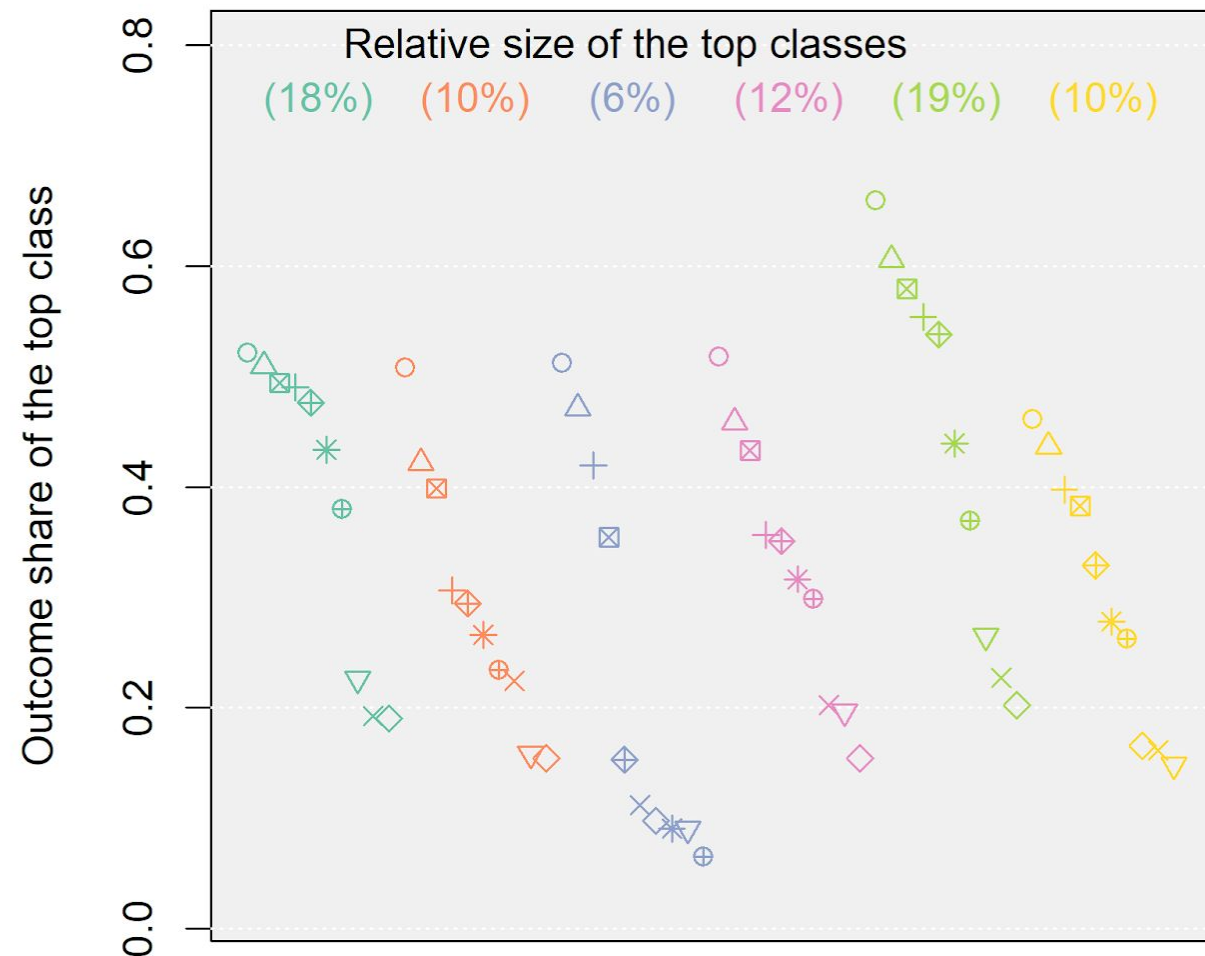
Macro field of science

- Agricultural Sciences
- Engineering and Technology
- Humanities
- Medical and Health Sciences
- Natural Sciences
- Social Sciences



Outcome

- Intl. publications
- △ Nat. publications
- + Nr. coauthored papers
- × Intl. moves
- ◇ Nat. moves
- ▽ Nr. of organizations
- ⊠ Total citations
- * Frac. publications
- ⊞ Nr. of publications
- ⊕ Nr. of first author pubs.



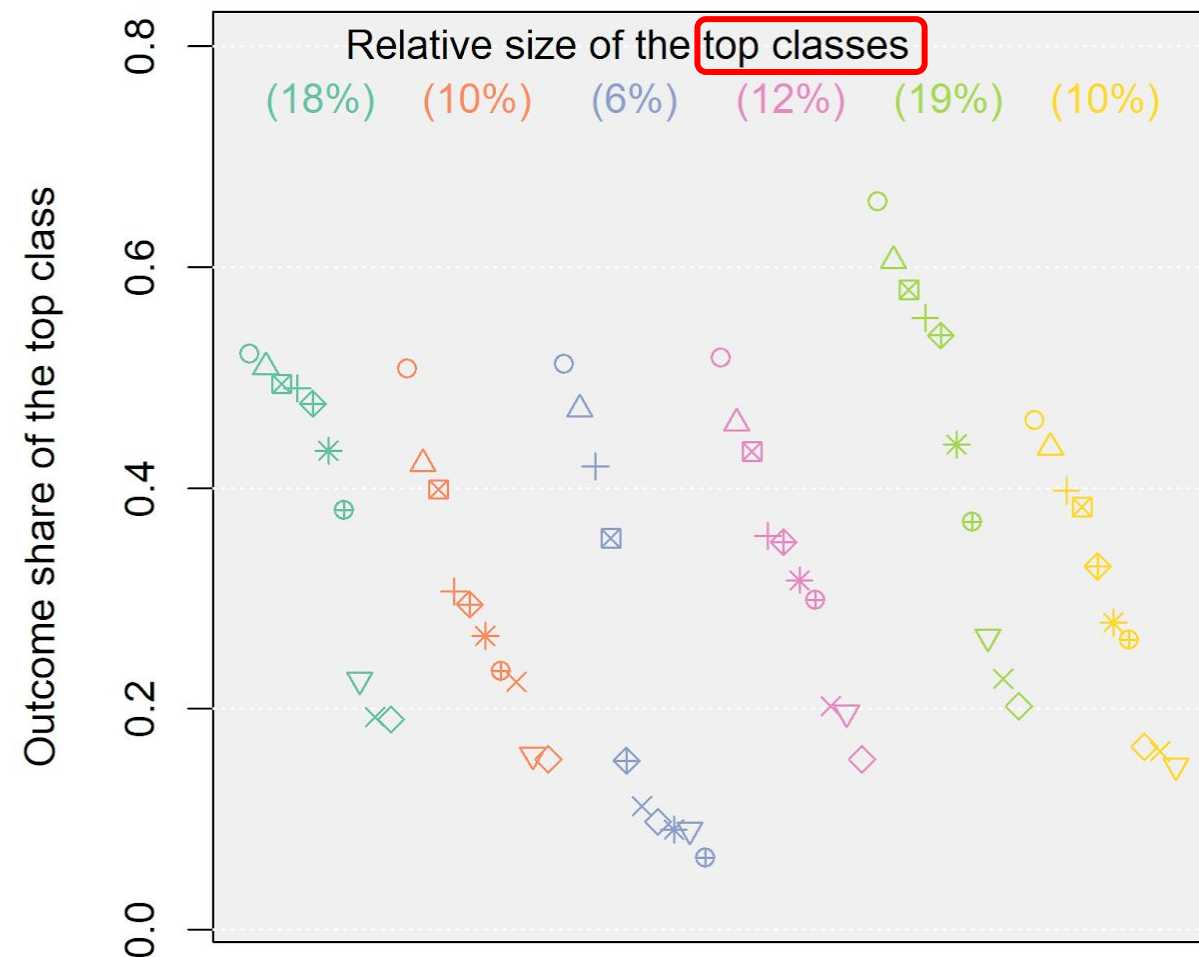
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- ⬢ Nr. of publications
- ⊕ Nr. of first author pubs.





We explored the hierarchical structure based on 12 variables.

How about collaboration networks?

Who co-authors publications with whom?

Do those in the top (most successful) collaborate only within their group?

Scopus co-authorship (bipartite / two-mode) network



Full network:

Vertices (nodes): **36,686,692**

Edges (ties): **93,102,217**

Giant component

(used for community detection):

Vertices (nodes): **34,768,821**

Edges (ties): **91,737,606**

CD using Leidenalg in Python

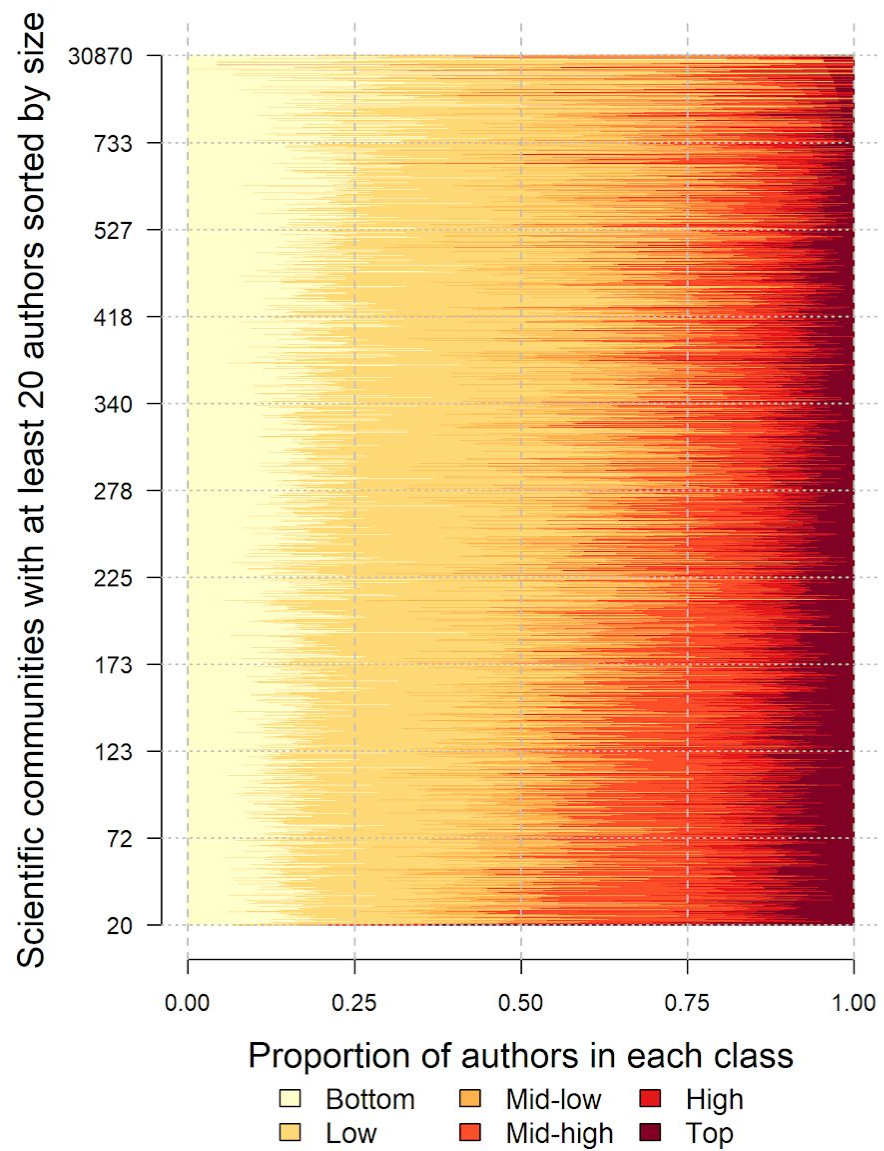
with CPM bipartite

18 different Gamma for resolution parameter (e.g., G-comp's density)

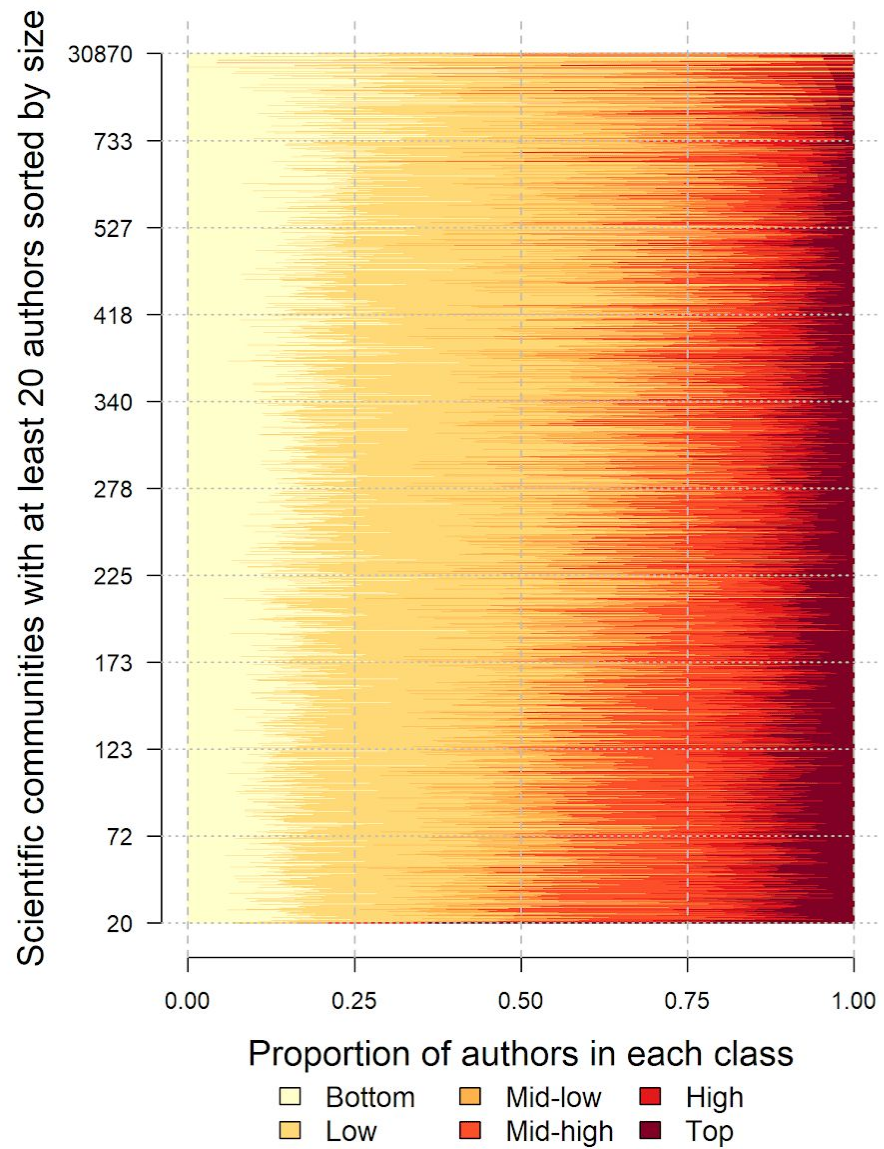
Robustness of results with Networkit in Python, using default, parallel Louvain, parallel label propagation (parallel leiden did not run). Disproportionately larger share of specific **countries** or **disciplines**? No!

Infomap (igraph) and BimLPA (CDlib) failed/or kept running (past 2 weeks)

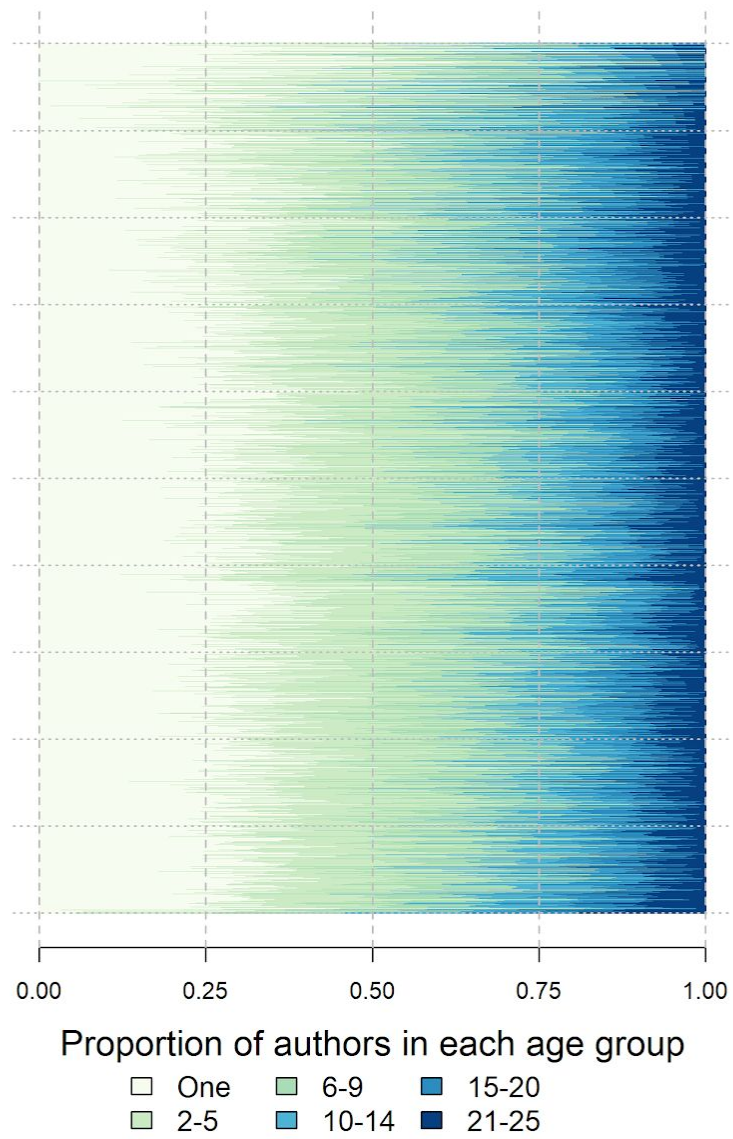
Panel A: Scientific communities' bibliometric stratification

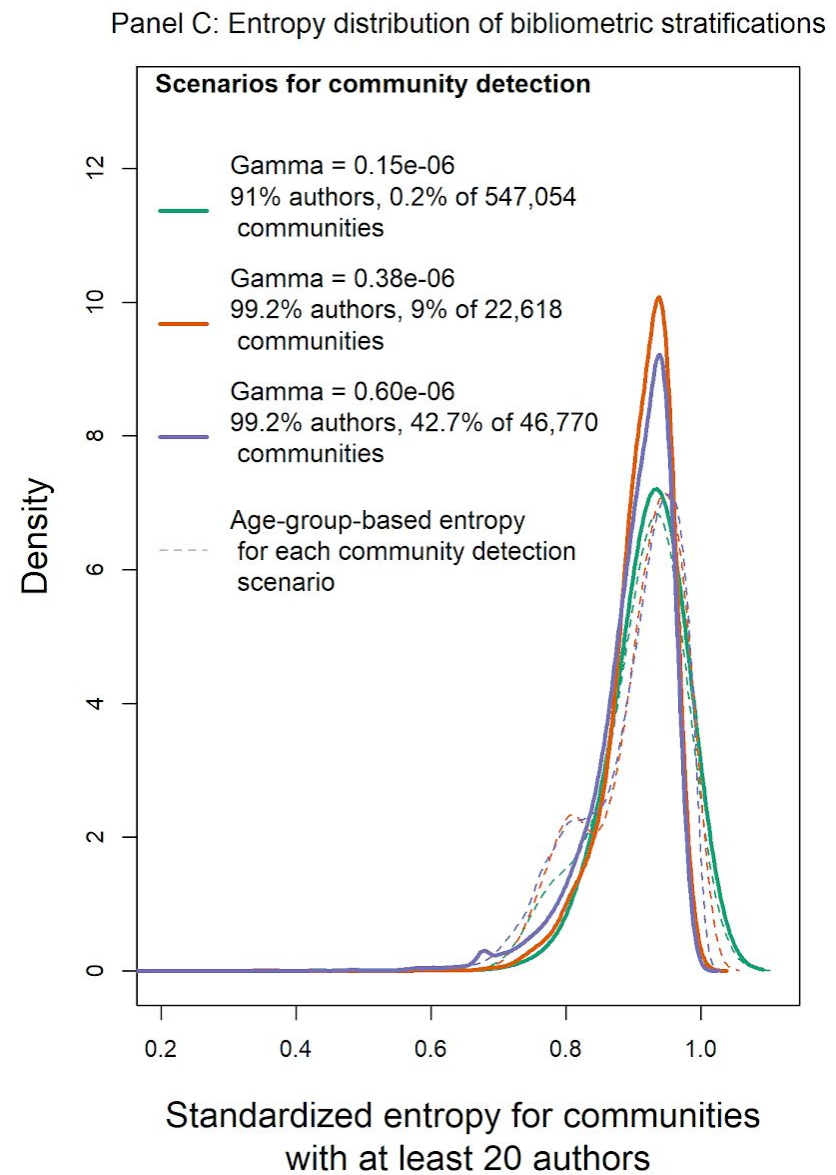
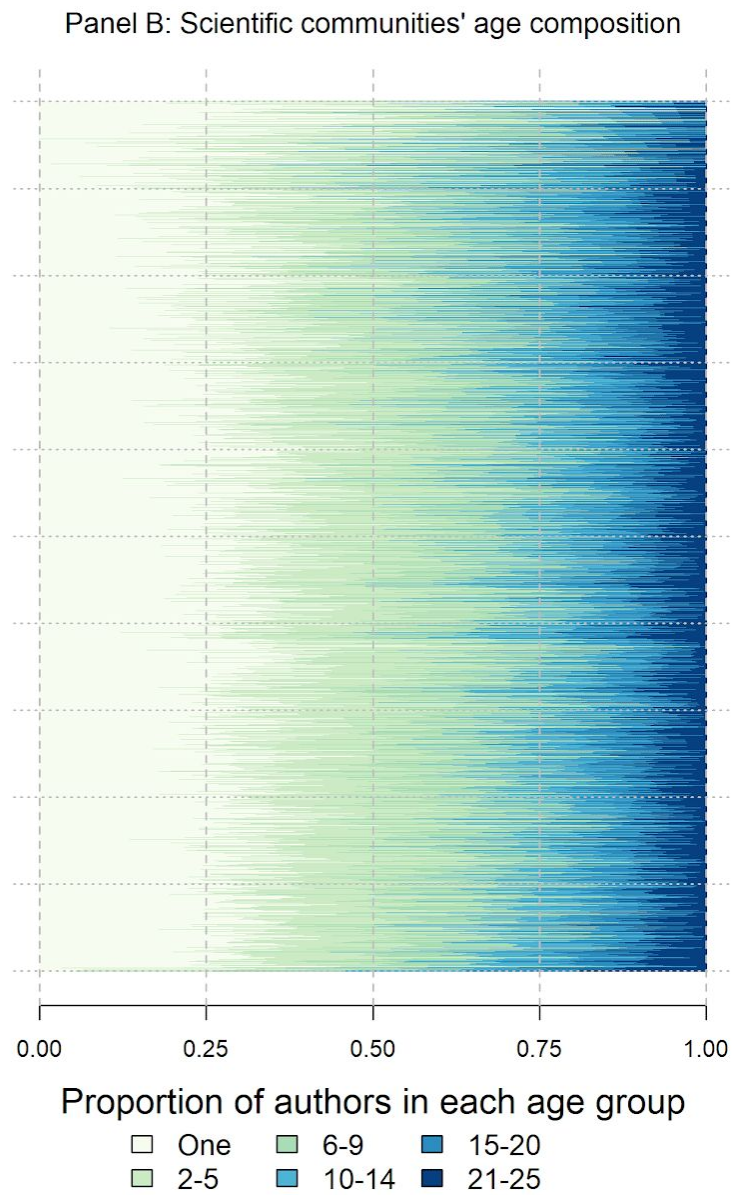
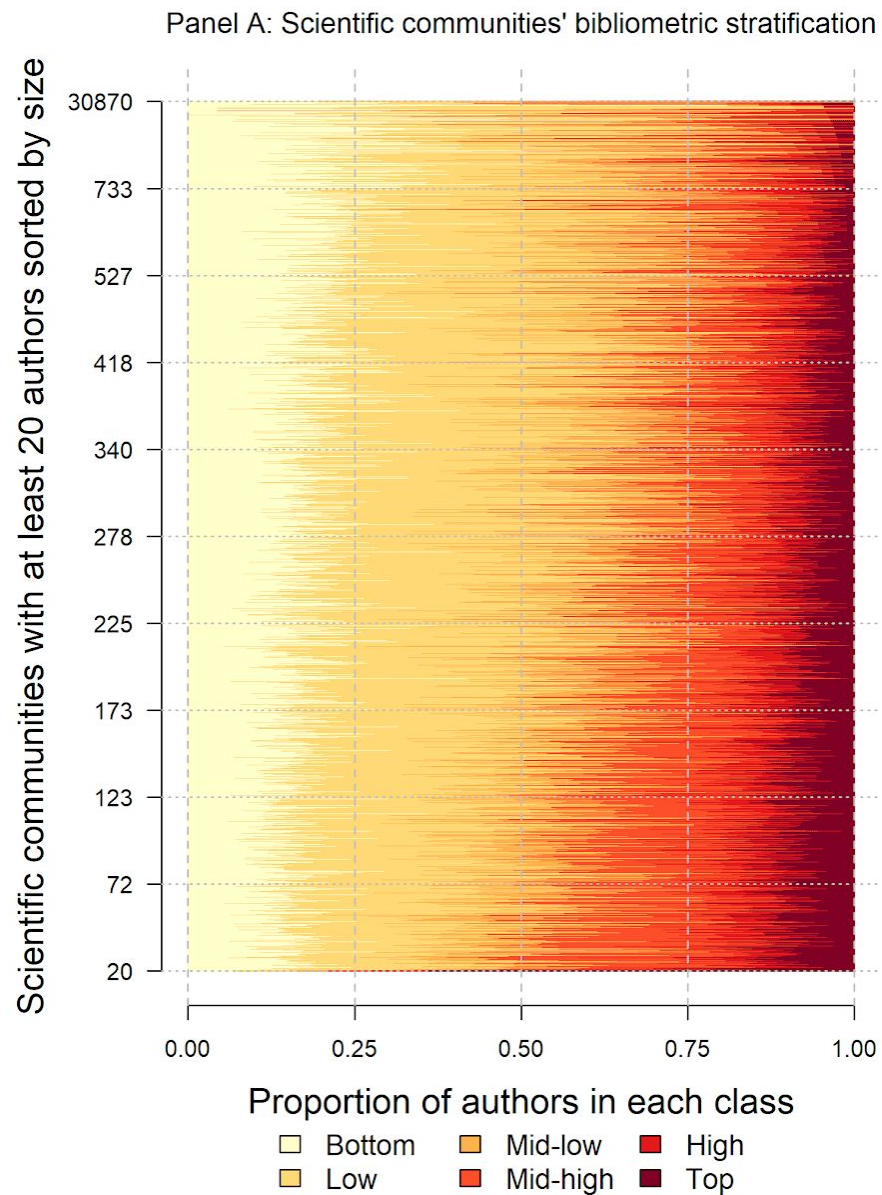


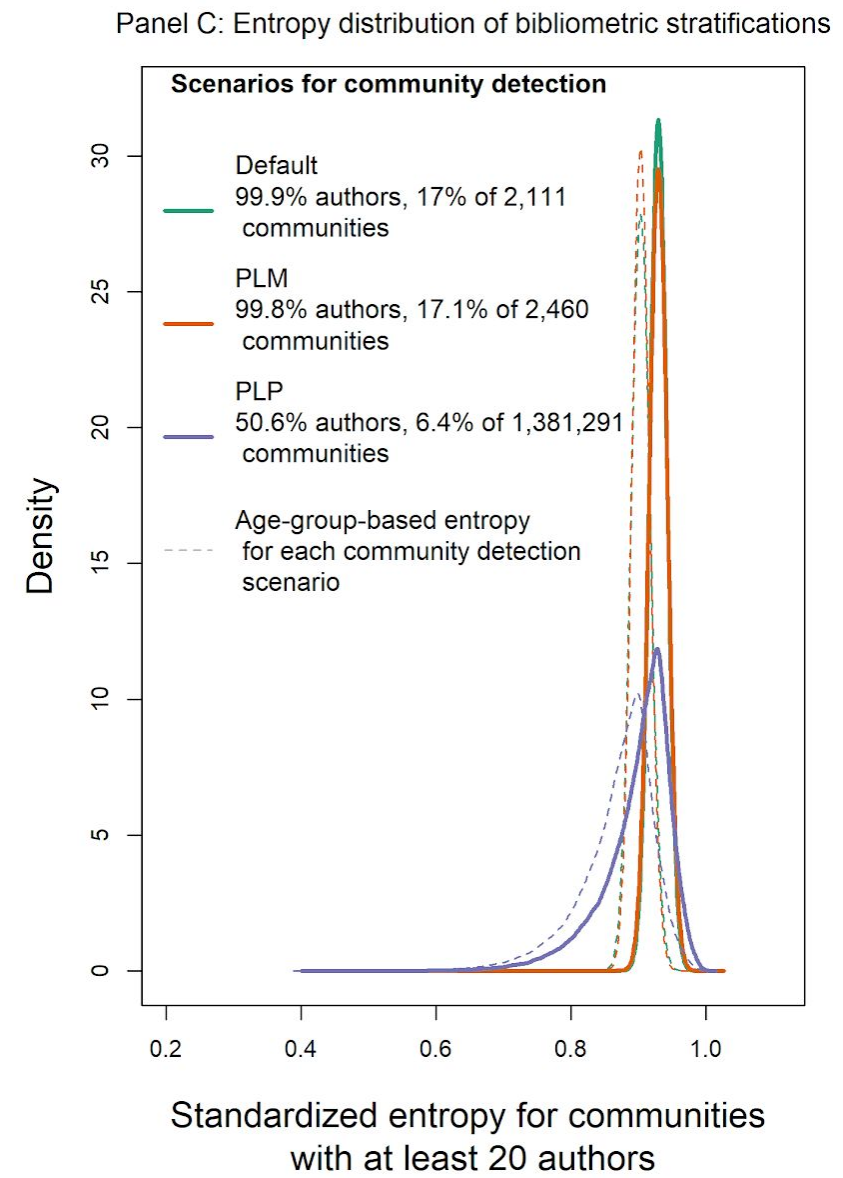
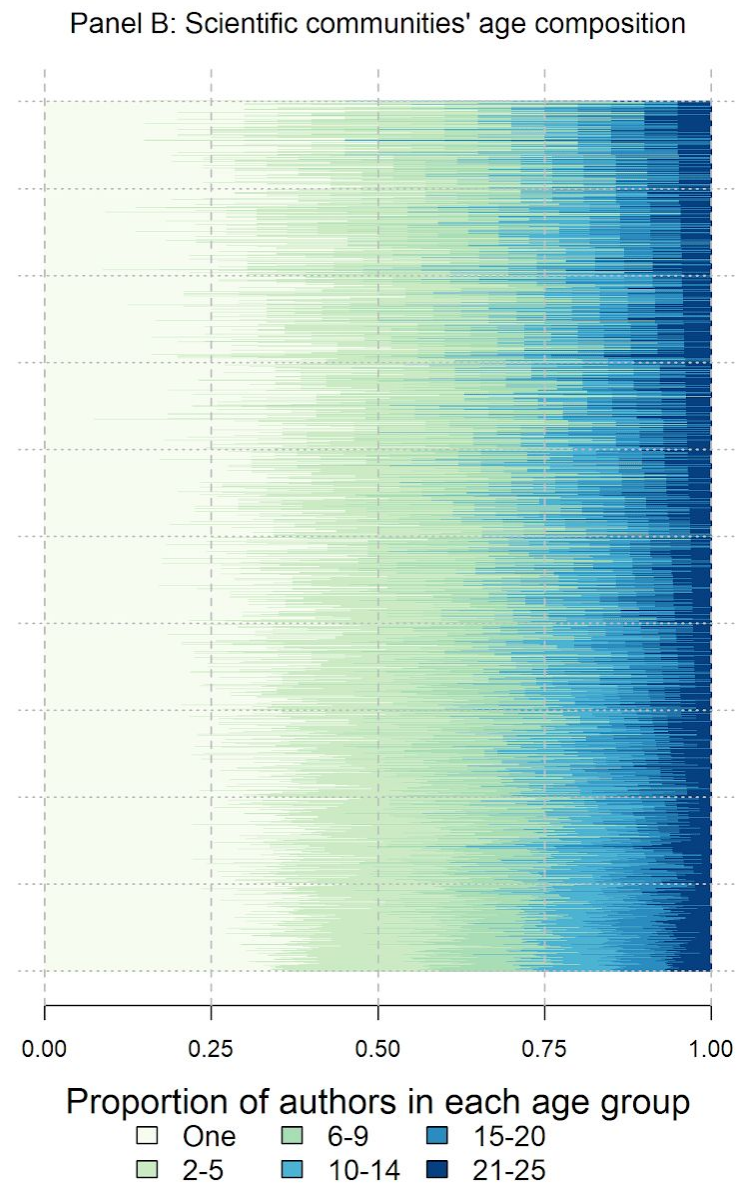
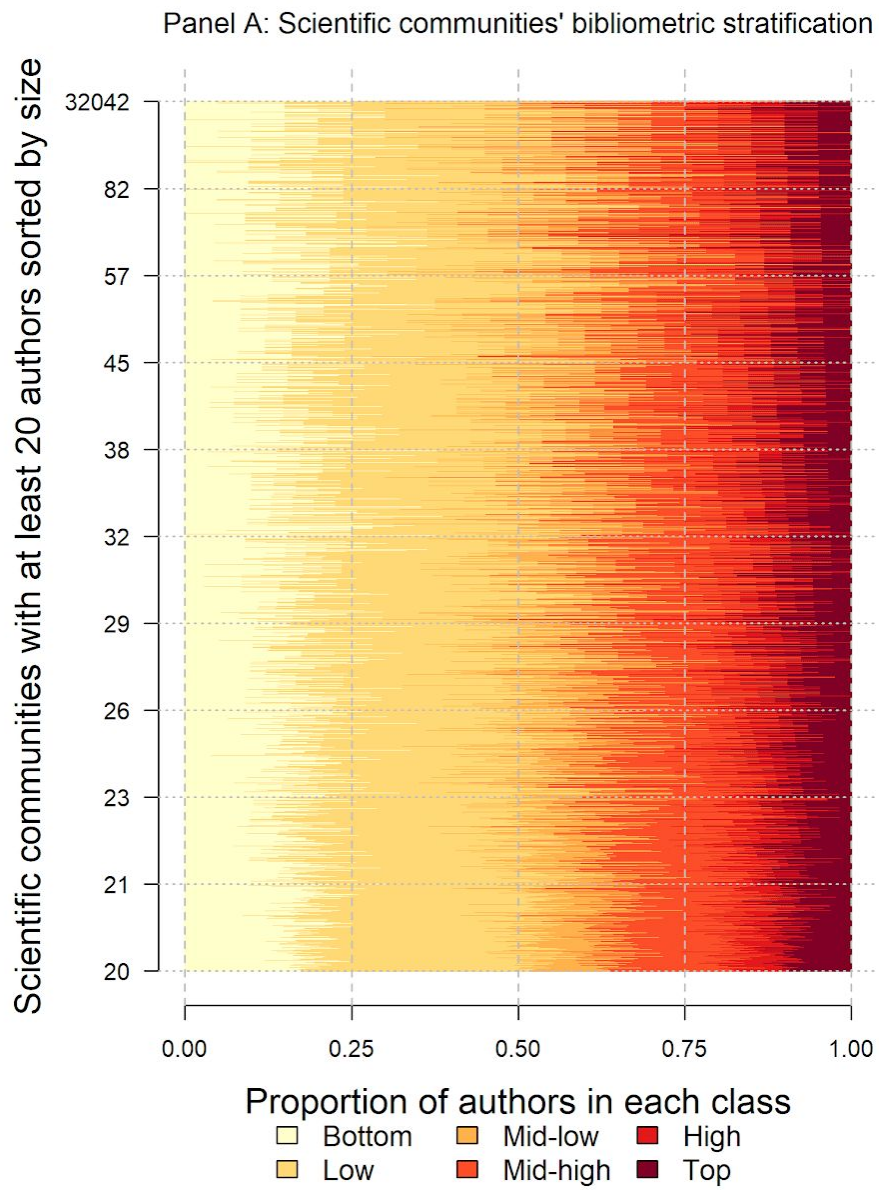
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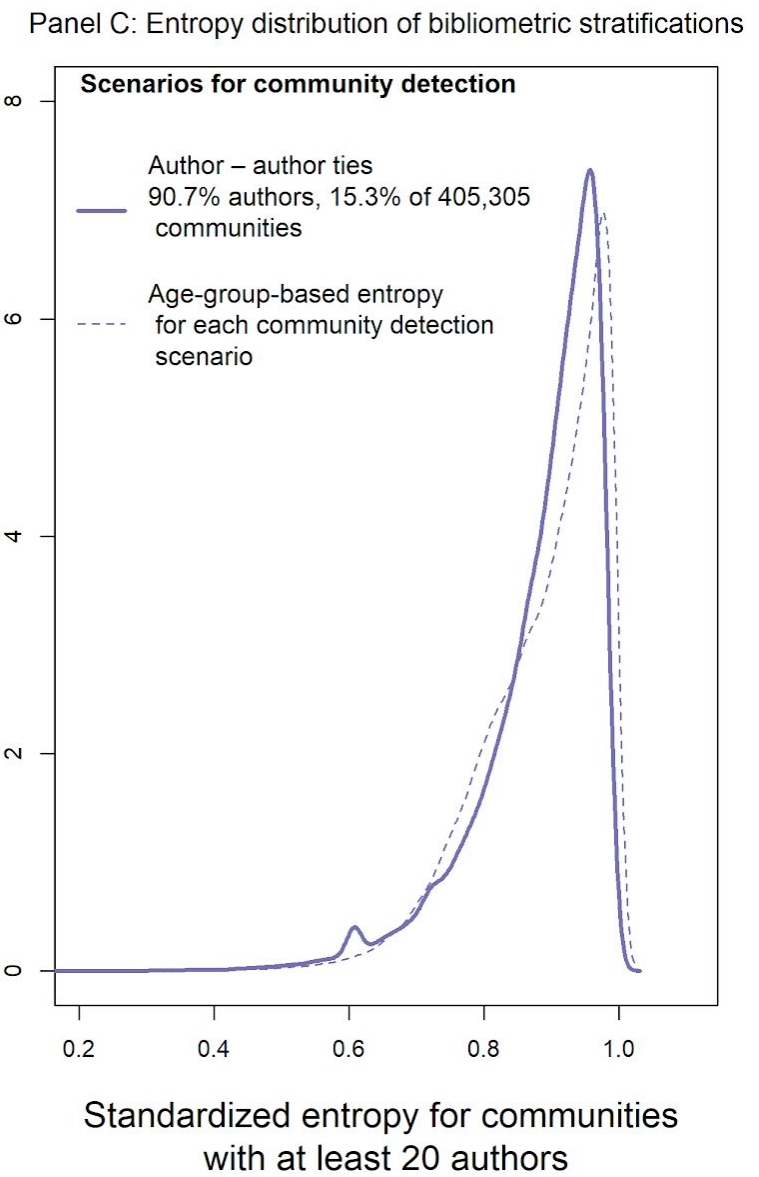
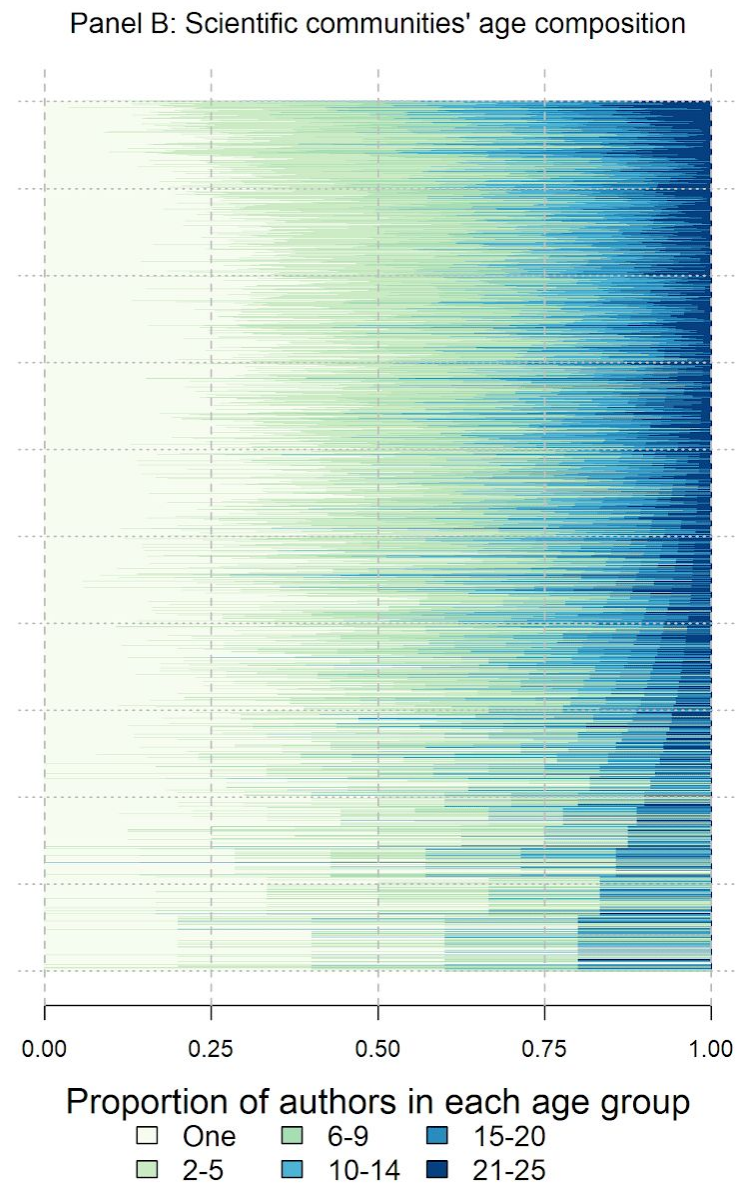
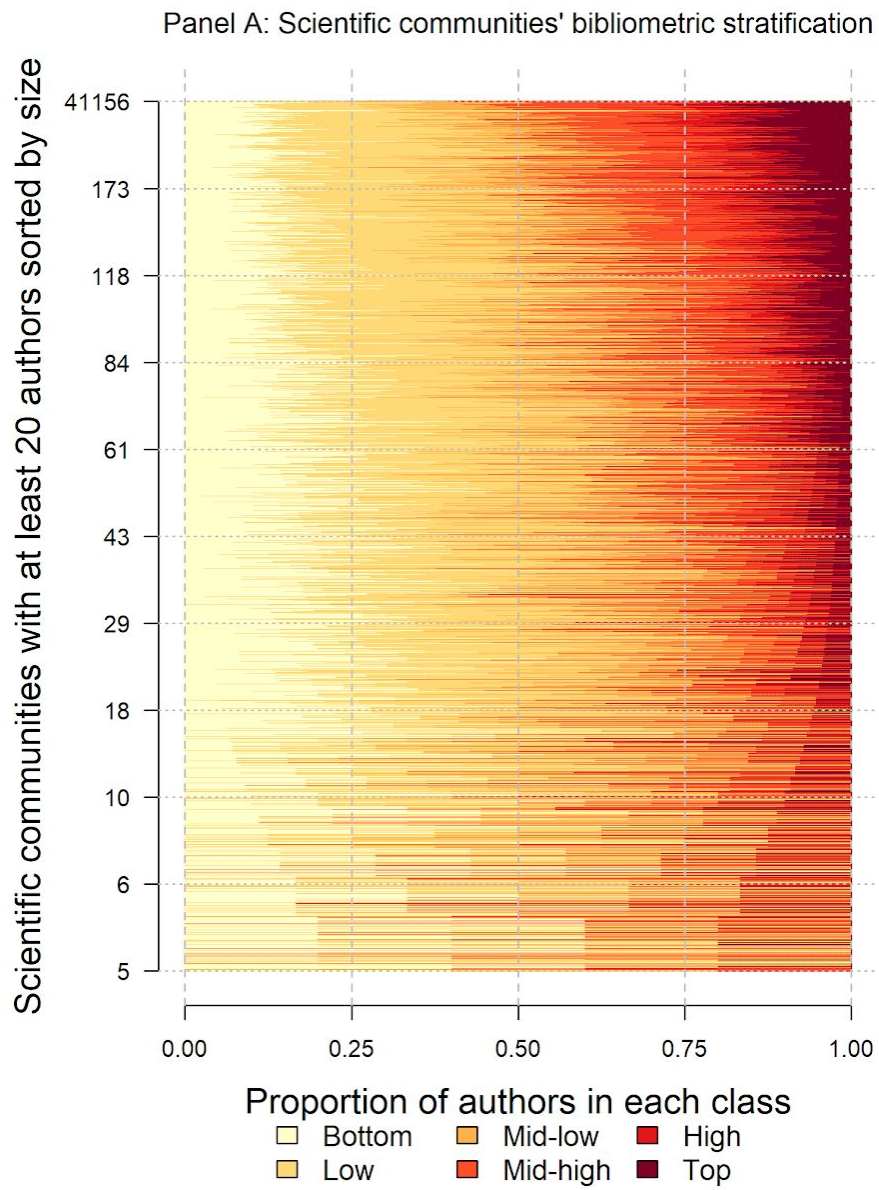
Panel B: Scientific communities' age composition





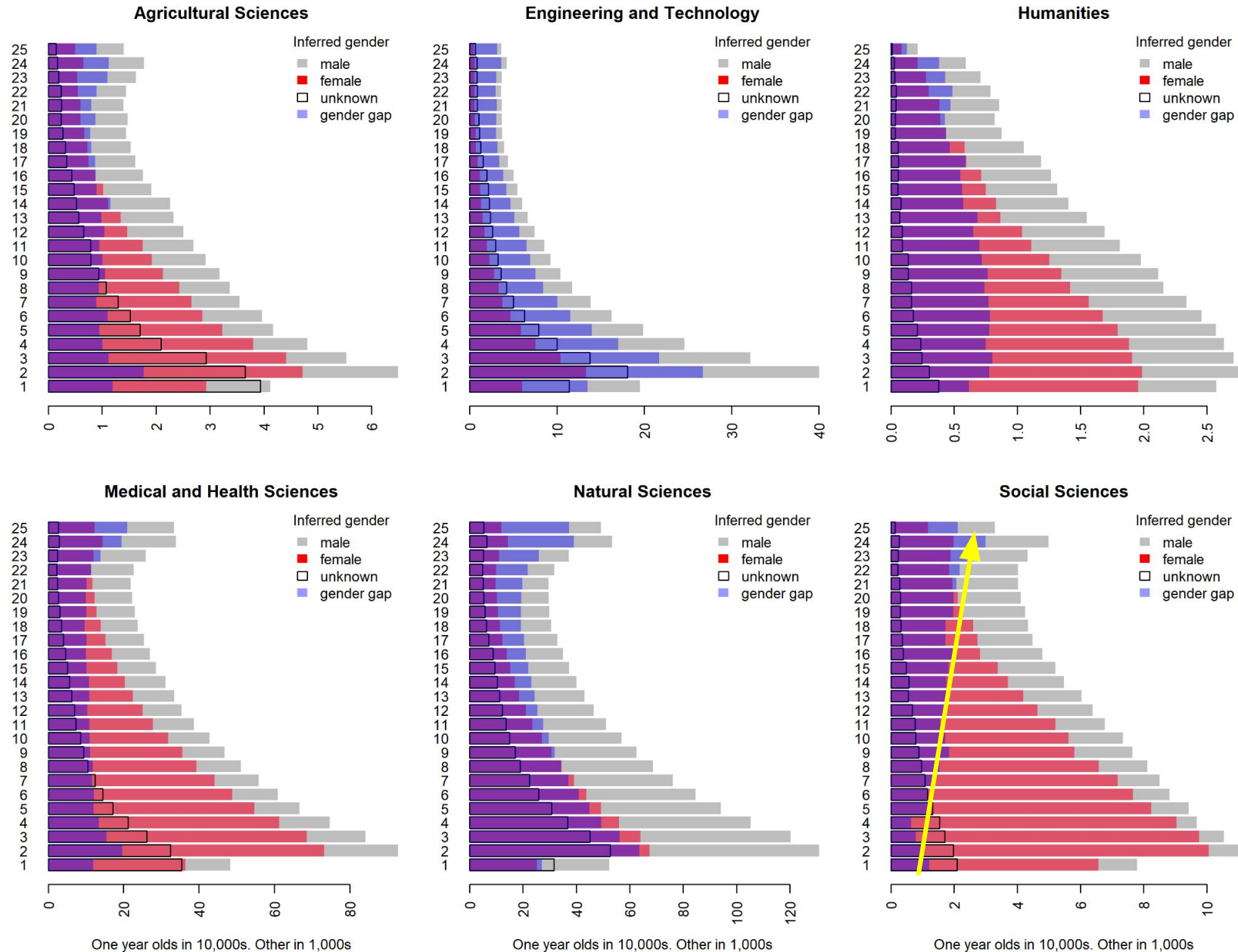


Robustness analysis: Using Networkit, 3 CD algorithms, default, parallel Louvain, parallel label propagation

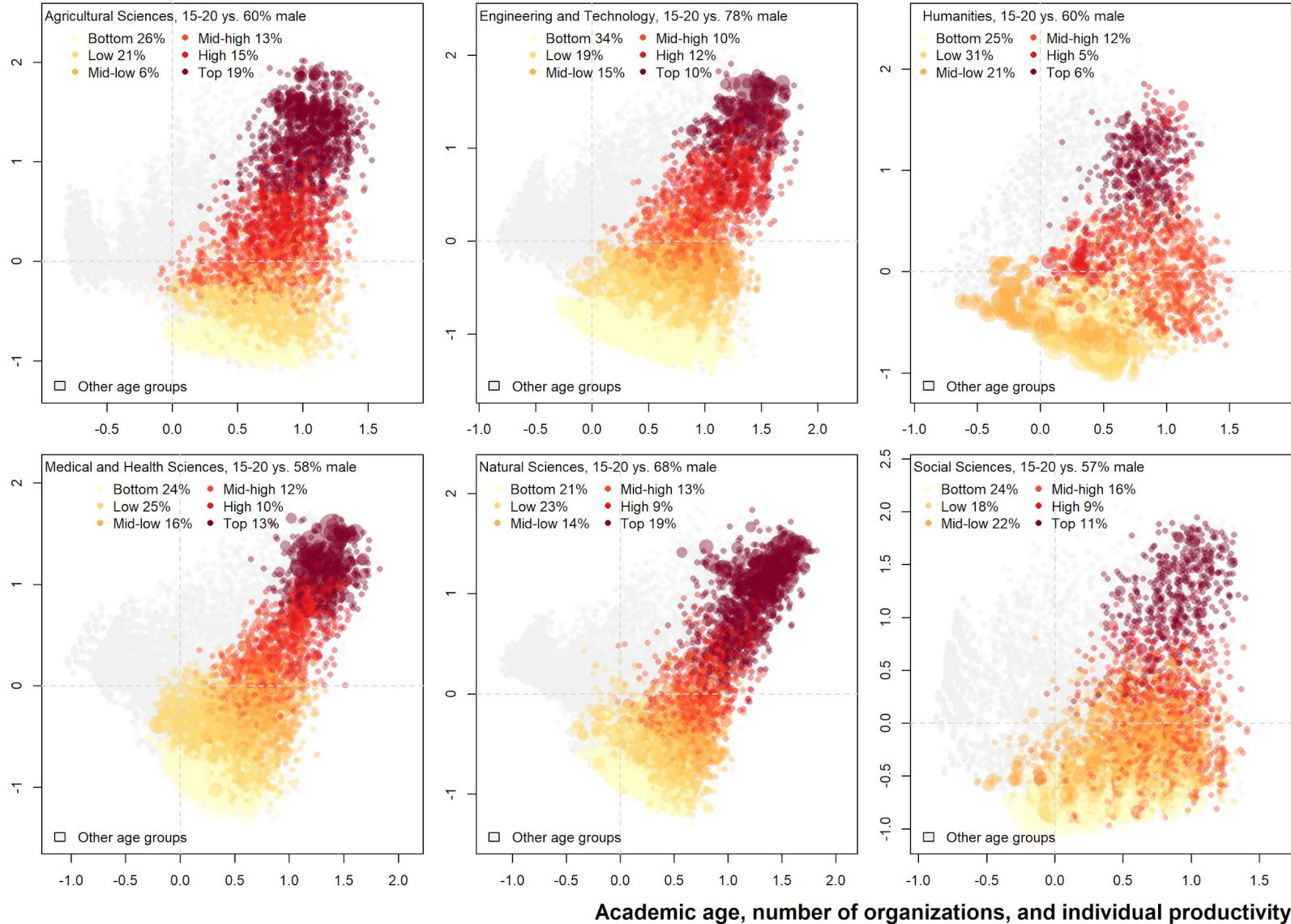


Robustness analysis: Using one mode projection of the network and Leiden algorithm (despite its cons)

How about gender's effect?

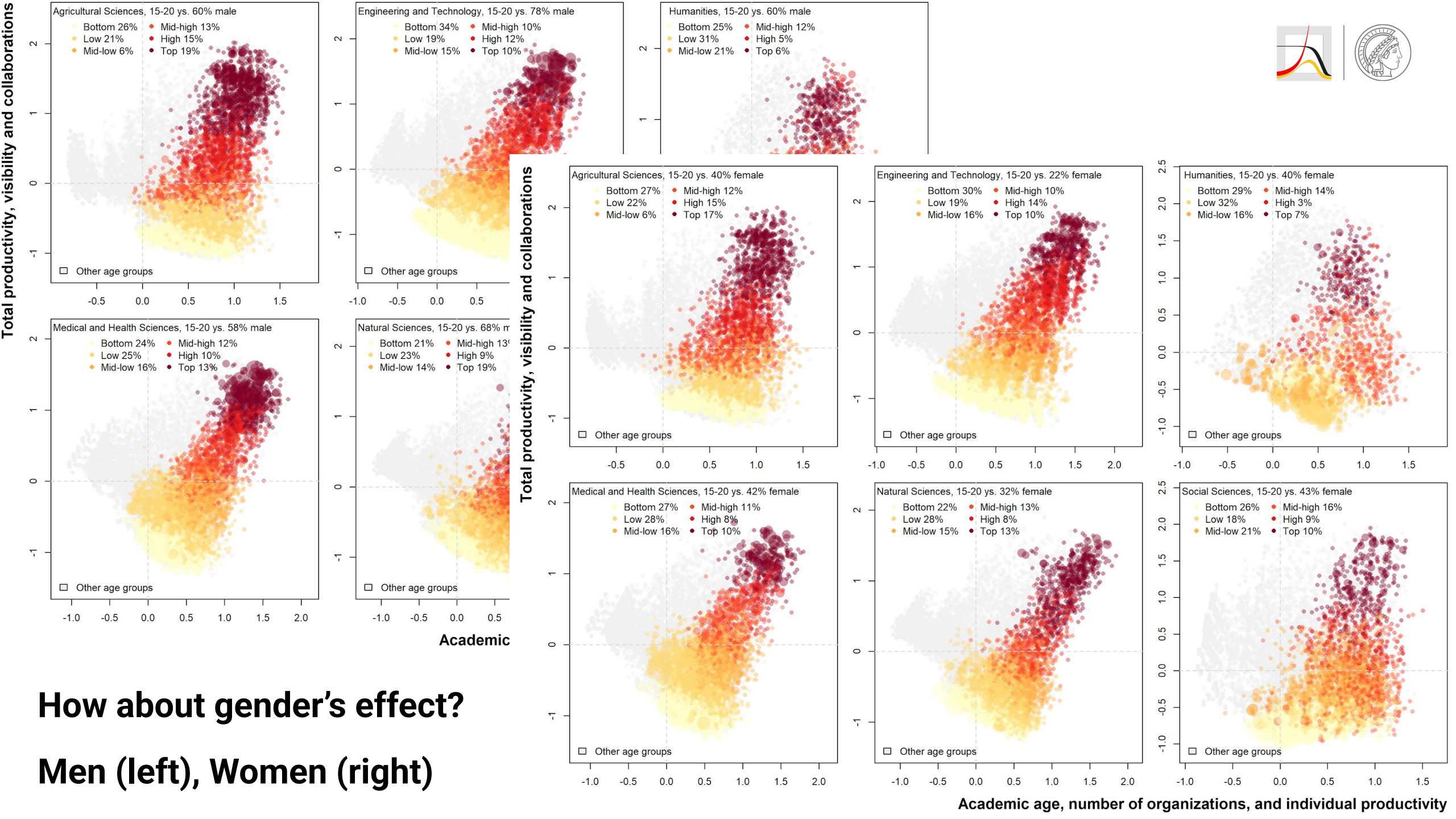


- Underrepresentation of women in all disciplines, especially in STEM.
- Increasing gender gap with academic age, very marked in the **Social Sciences** but also in other disciplines



How about gender's effect?

Men (left), Women (right)



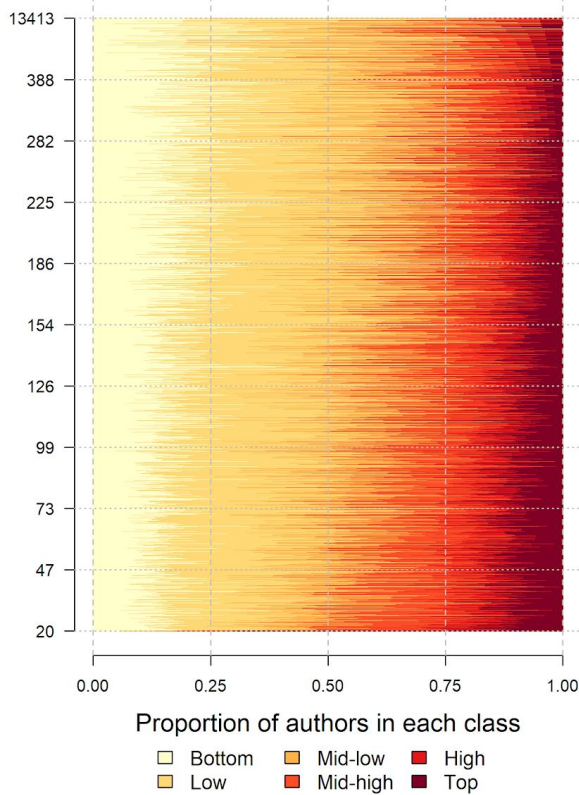
Percentage distribution across bibliometric classes by inferred gender for selected countries in the Social Sciences (N = 448,446)

Country	Bottom	Low	Mid-low	Mid-high	High	Top	Total	%
Brazil fem	60	7	18	7	2	7	100	
Brazil mal	44	8	27	16	1	3	100	
Brazil unkn	20	20	40	0	0	20	100	
Total							141	100
Germany fem	26	16	19	9	2	28	100	
Germany mal	24	15	20	15	3	22	100	
Germany unkn	15	8	23	31	8	15	100	
Total							845	100
South Korea fem	26	5	26	21	2	19	100	
South Korea mal	38	11	25	7	7	13	100	
South Korea unkn	44	6	30	13	4	4	100	
Total							200	100
Spain fem	34	4	26	18	7	12	100	
Spain mal	30	6	24	20	5	15	100	
Spain unkn	22	11	24	32	8	3	100	
Total								100
United States fem	31	11	24	13	4	17	100	
United States mal	26	11	25	15	4	19	100	
United States unkn	29	8	24	15	8	16	100	
Total							13,560	100

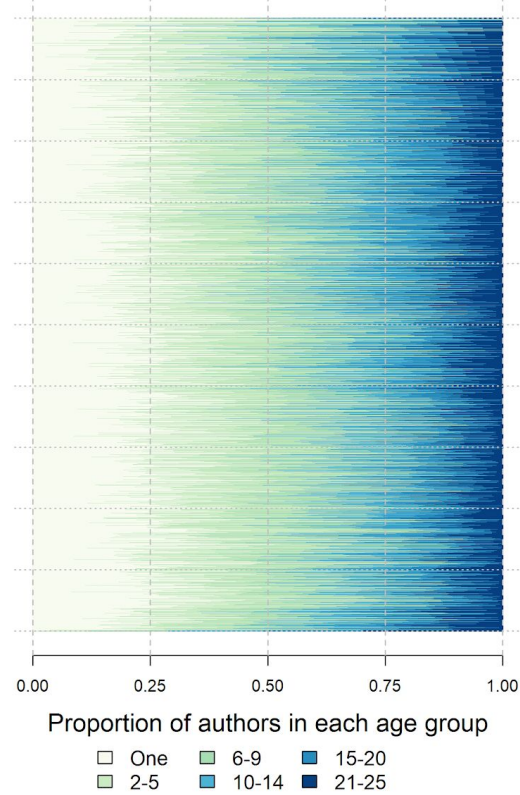
- Underrepresentation of women, particularly in Germany
- Differences in the algorithm's capacity to inferred gender, e.g., South Korea
- Similar bibliometric class structure between genders

Scientific communities with at least 20 authors sorted by size

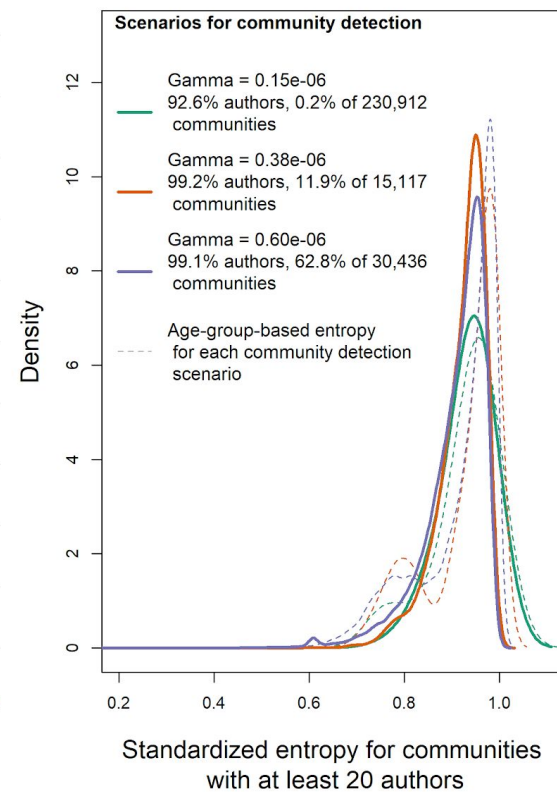
Panel A: Scientific communities' bibliometric stratification



Panel B: Scientific communities' age composition



Panel C: Entropy distribution of bibliometric stratifications

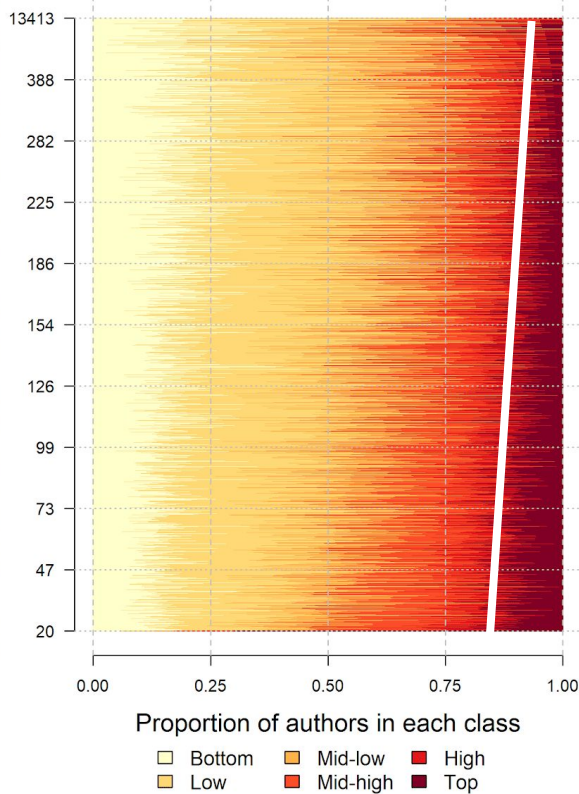


How about gender's effect?

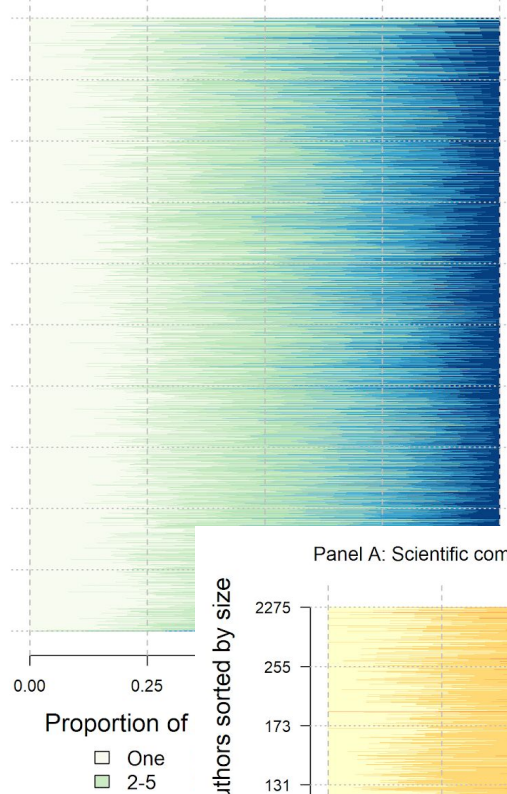
Men (left), Women (right)

Scientific communities with at least 20 authors sorted by size

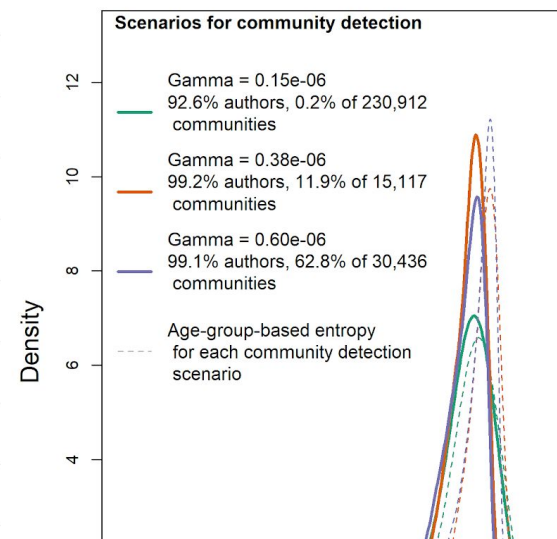
Panel A: Scientific communities' bibliometric stratification



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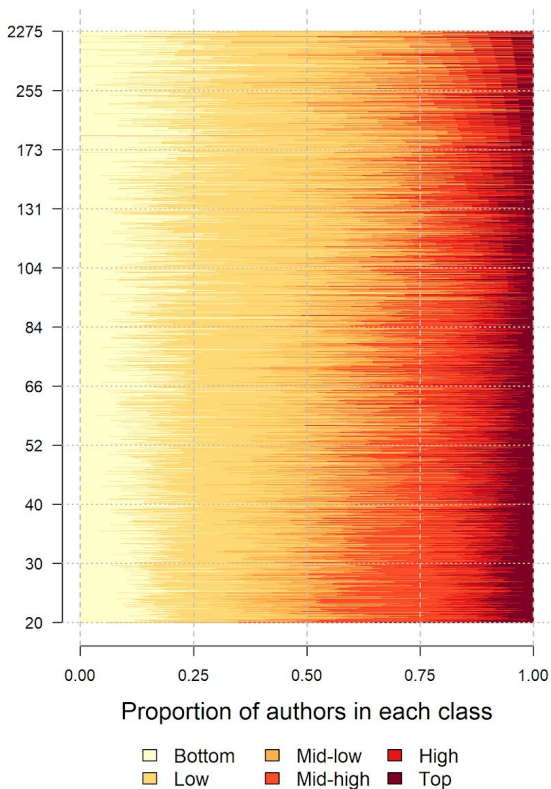


How about gender's effect?

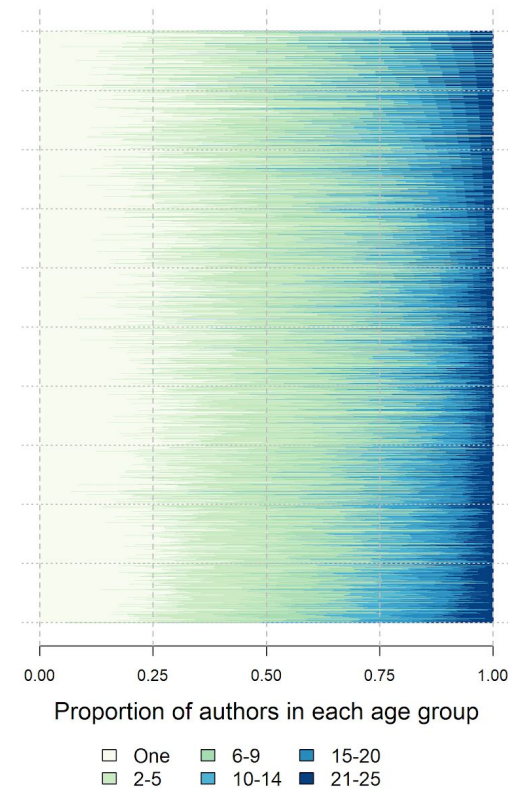
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Scientific communities with at least 20 authors sorted by size

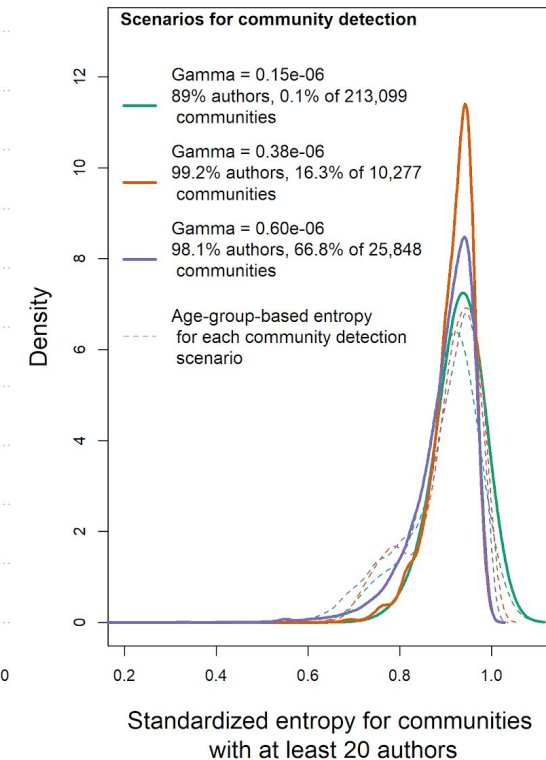
Panel A: Scientific communities' bibliometric stratification



Panel B: Scientific communities' age composition



Panel C: Entropy distribution of bibliometric stratifications



Conclusions



- A **quantitative** assessment of the **global inequalities in science**,
- Using **bibliometric** data, across **fields of science** and **within** research communities.
- Claims on increased productivity, collaboration, and mobility are principally driven by a small fraction of influential scientists (top 10%). It is shocking not the **gender** issue.
- A **stratified system** exists, and it is as strong as stratification by **academic age**. (a small top class, and large middle and bottom classes)
- Those at the top **succeed by collaborating** with a varying group of authors from **other classes** and **age** groups.
- Nevertheless, they are **benefiting disproportionately** to a much higher degree from this collaboration and its outcome in form of **impact** and **citations**.
- **Supervisor-supervisee** relationships inherently have an **age structure**, (Our observation goes beyond that ($25\% < 5$ yrs))
- We propose considering academic age, career cohorts and composition of a multitude of bibliometric variables instead of solely relying on one-indicator explanations.

We thank **Cassidy R. Sugimoto** for helpful comments on an earlier version of this manuscript.



Thank you for your attention!

Feedback, comments, and questions are very welcome.

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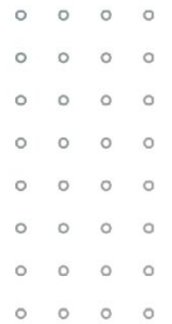
Working paper



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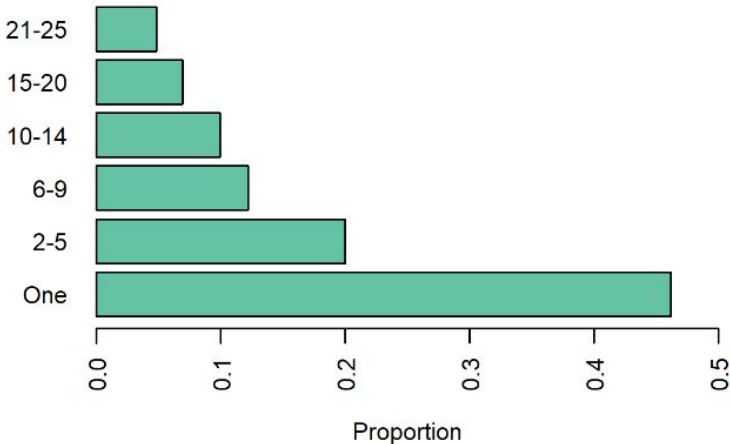
Additional slides for Q&A



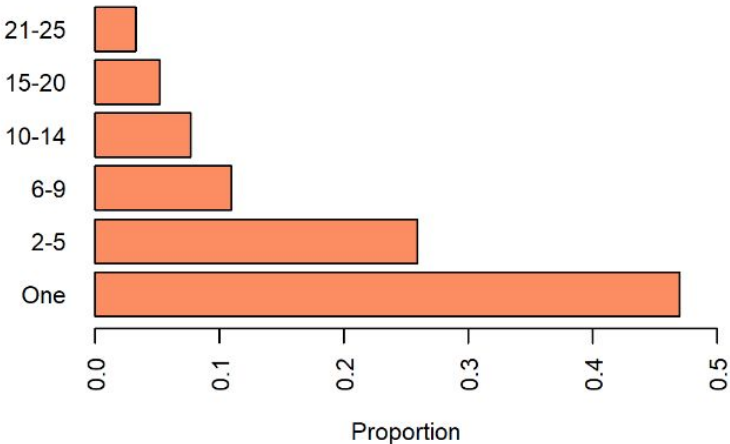
Academic age distribution of scholars in the analytical sample by OECD macro fields of science



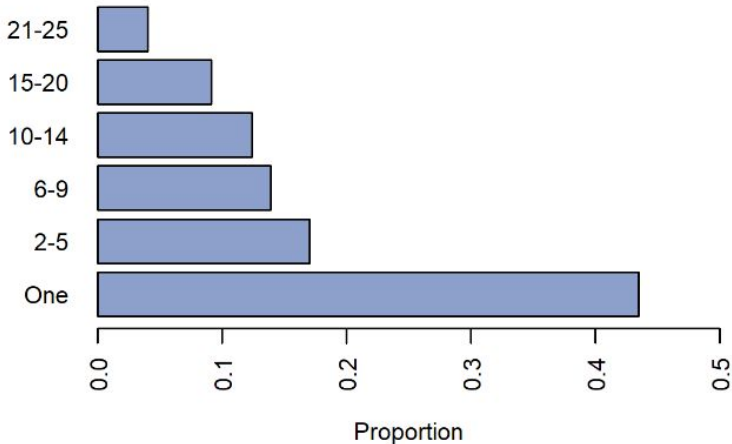
Agricultural Sciences



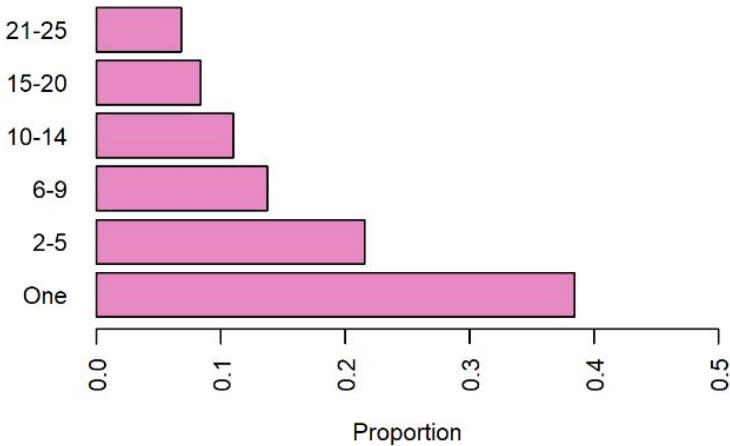
Engineering and Technology



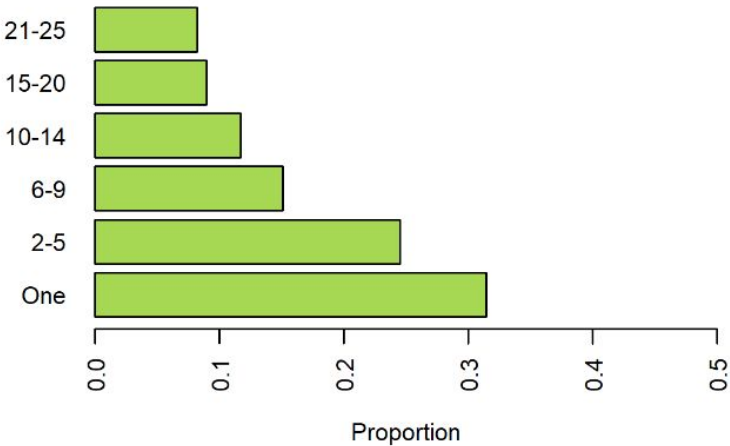
Humanities



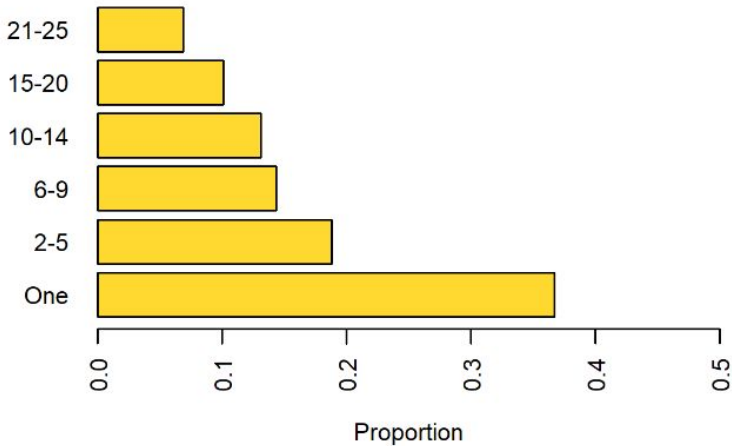
Medical and Health Sciences



Natural Sciences

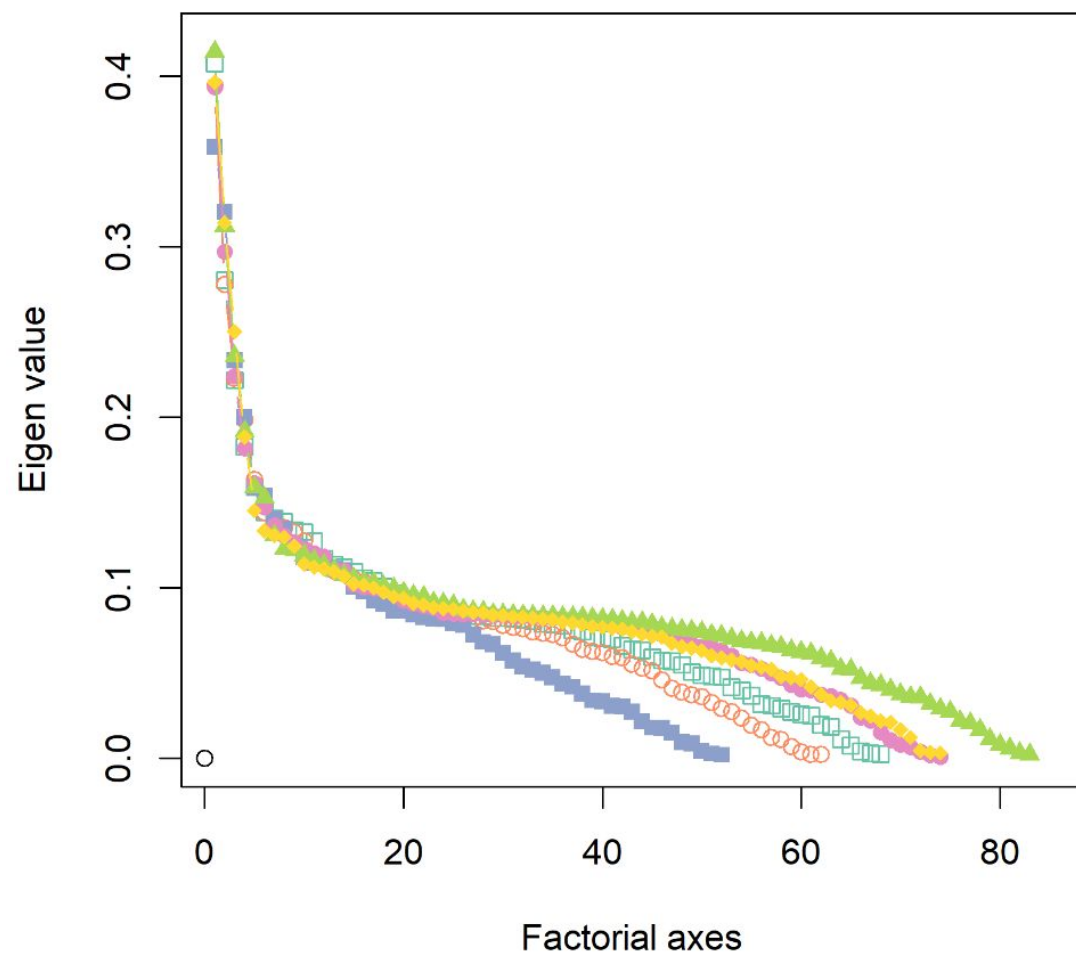


Social Sciences

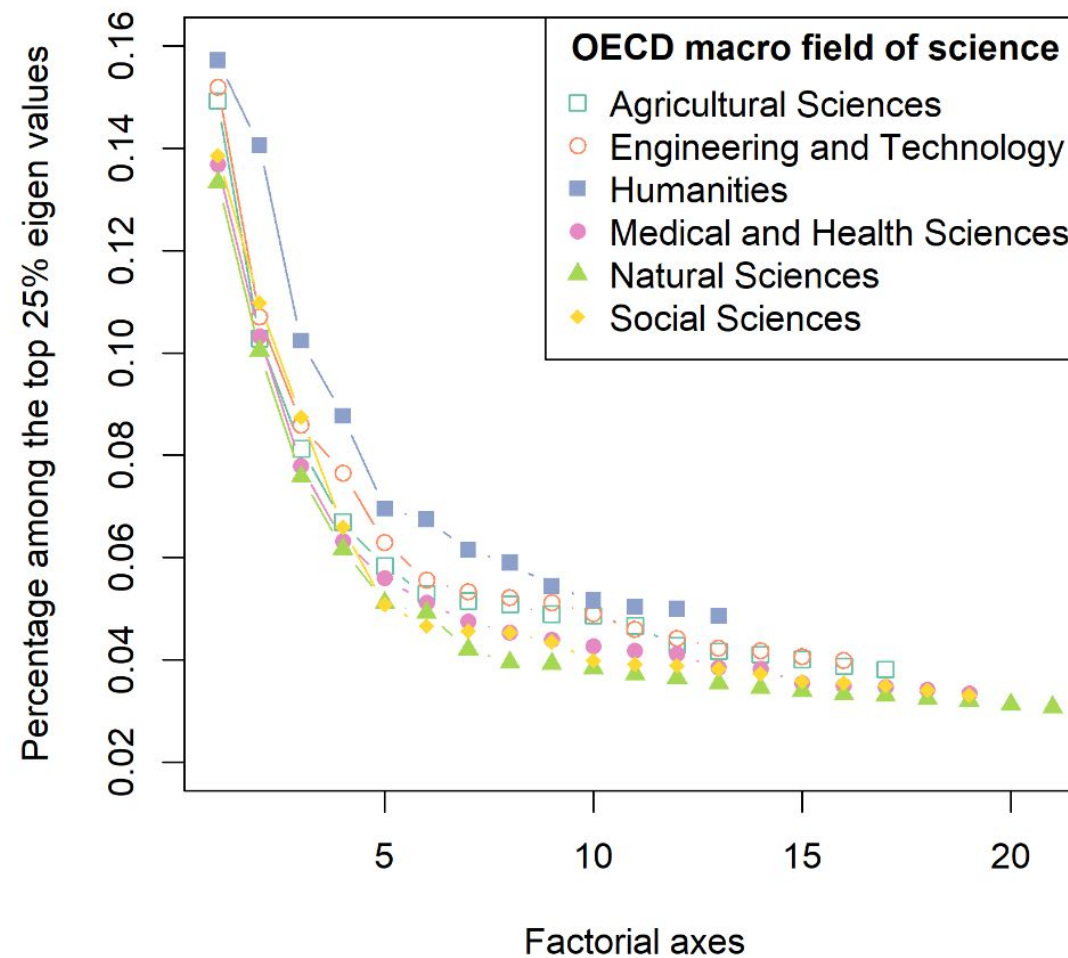




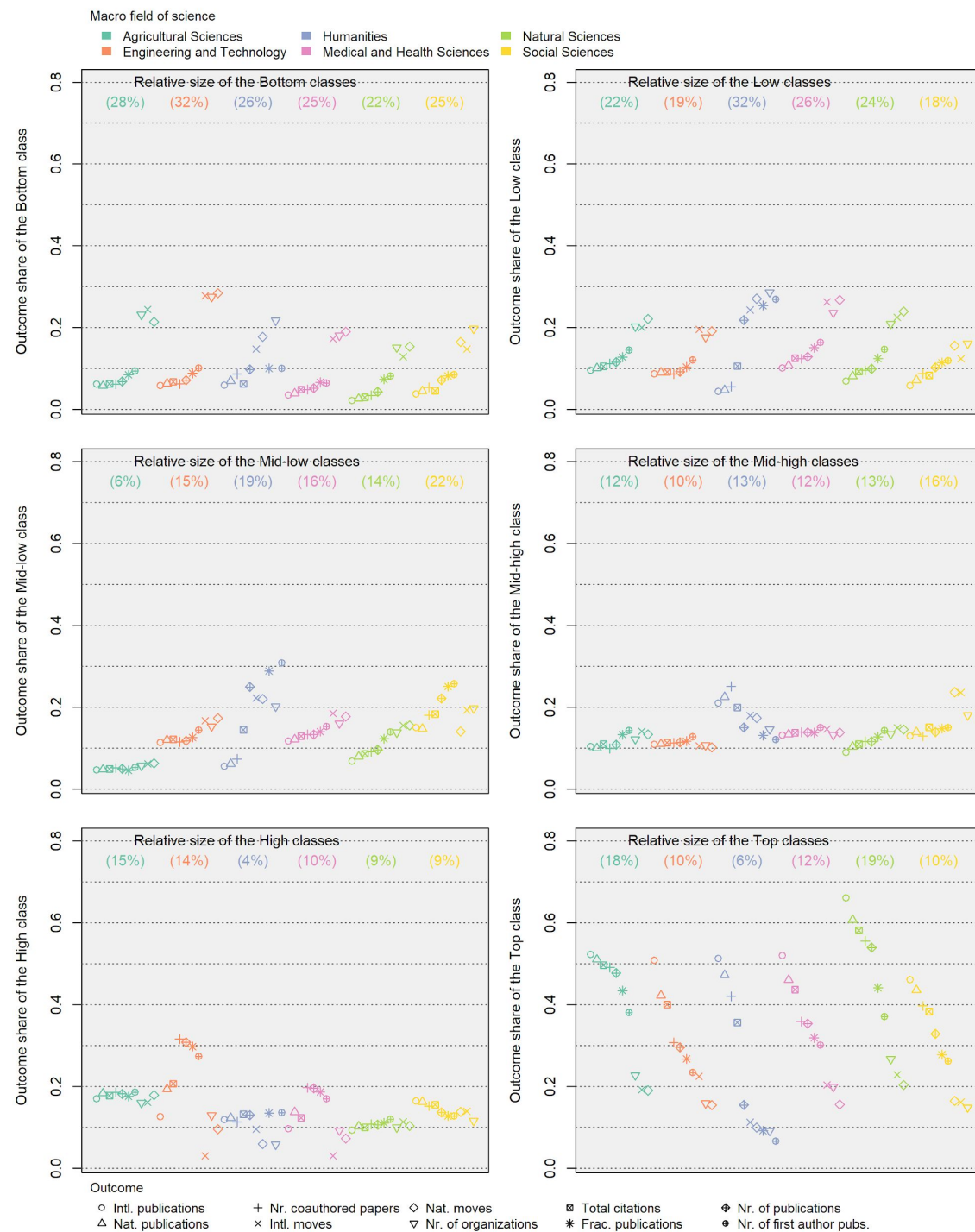
All factorial axes



Top 25% eigen values



Share contribution of
bibliometric classes to 10
academic performance
indicators by macro field
of science for 15 to 20
years old scholars.

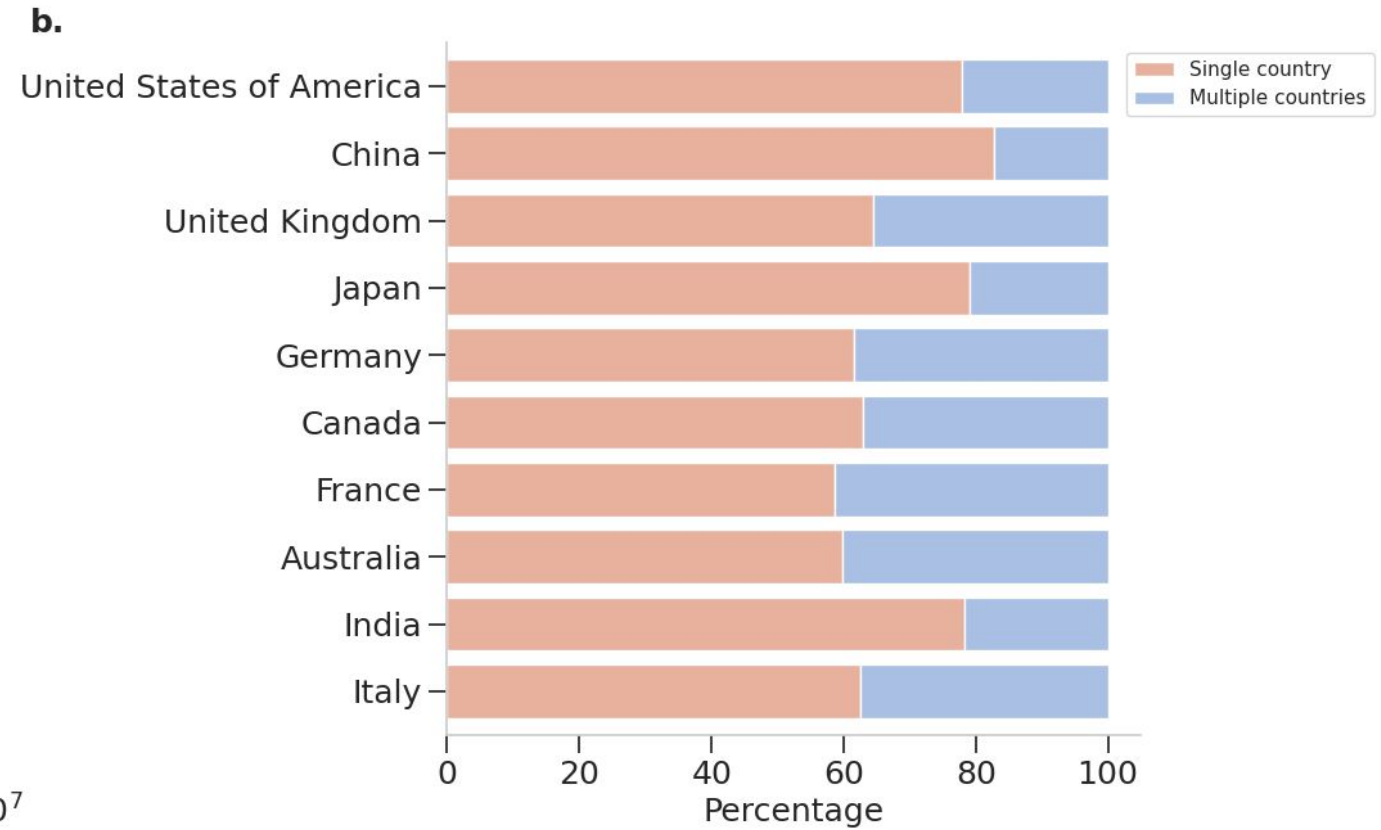
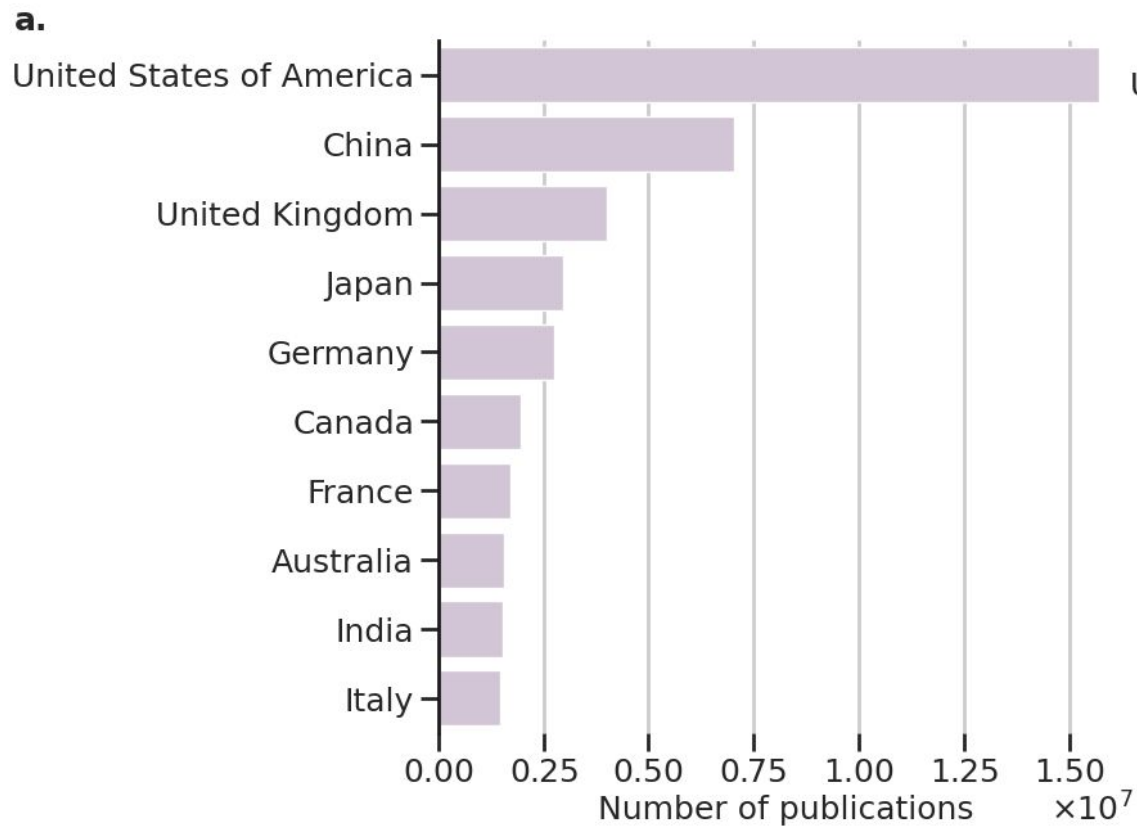


Absolute number of scholars by inferred gender for selected countries in the Social Sciences (N = 448,446)

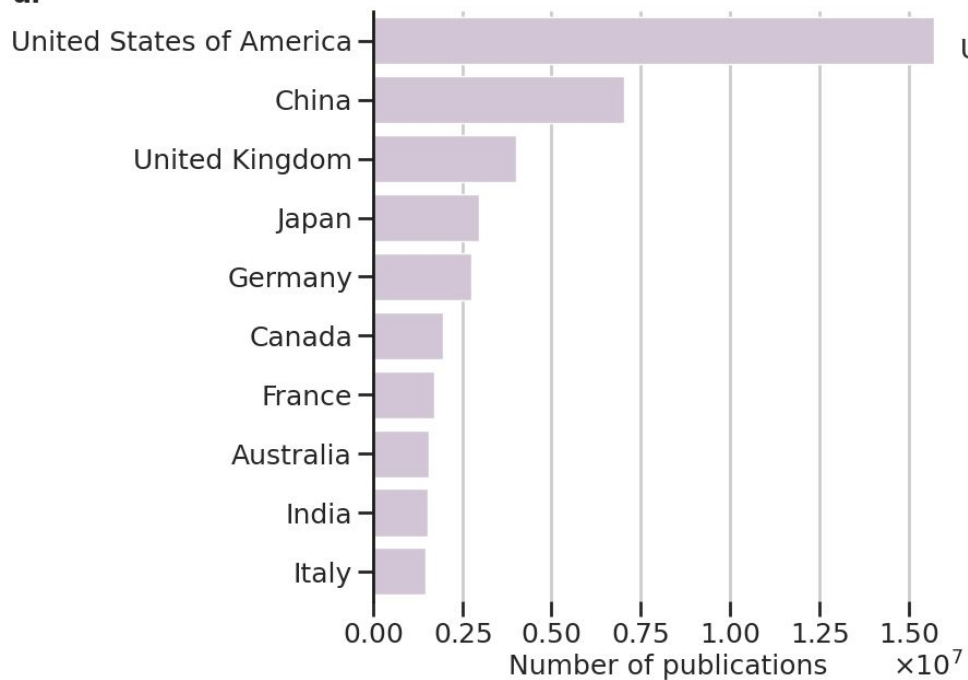
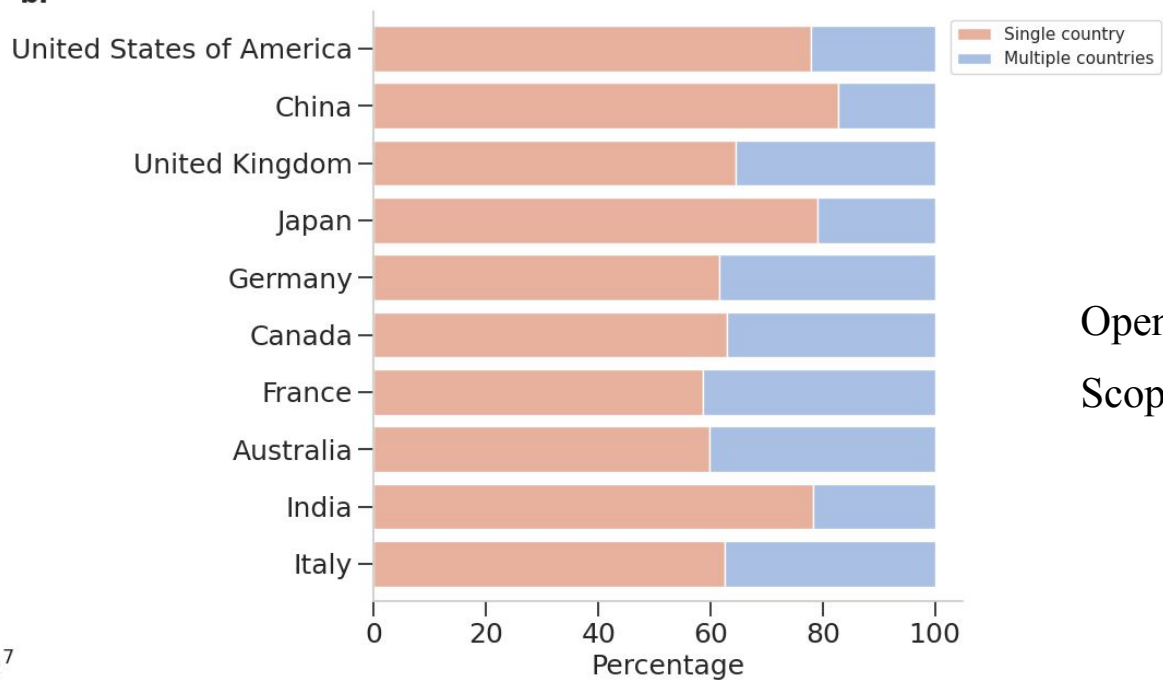
Country	Bottom	Low	Mid-low	Mid-high	High	Top	Total	%
Brazil fem	27	3	8	3	1	3	45	32
Brazil mal	40	7	25	15	1	3	91	65
Brazil unkn	1	1	2	-	-	1	5	4
Total							141	100
Germany fem	35	21	26	12	3	37	134	16
Germany mal	171	105	140	107	23	152	698	83
Germany unkn	2	1	3	4	1	2	13	2
Total							845	100
South Korea fem	11	2	11	9	1	8	42	21
South Korea mal	40	11	26	7	7	13	104	52
South Korea unkn	24	3	16	7	2	2	54	27
Total							200	100
Spain fem	99	11	75	52	19	34	290	34
Spain mal	157	32	127	106	27	82	531	62
Spain unkn	8	4	9	12	3	1	37	4
Total								100
United States fem	1,287	469	1,023	537	164	707	4,187	31
United States mal	2,282	1,018	2,217	1,345	370	1,685	8,917	66
United States unkn	134	38	108	67	35	74	456	3
Total							13,560	100

- Underrepresentation of women, particularly in Germany
- Differences in the algorithm's capacity to inferred gender, e.g., South Korea
- Similar bibliometric class structure between genders

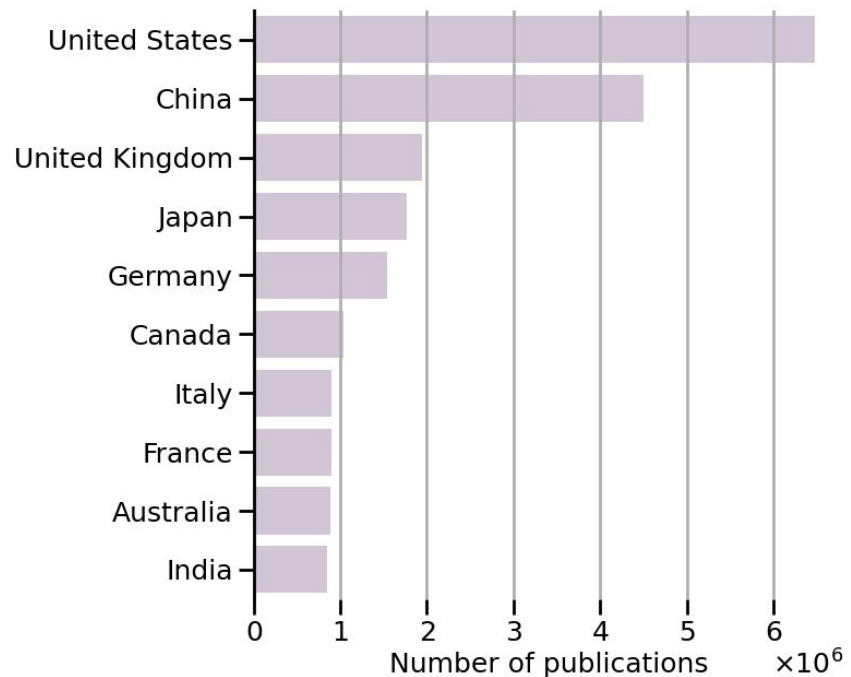
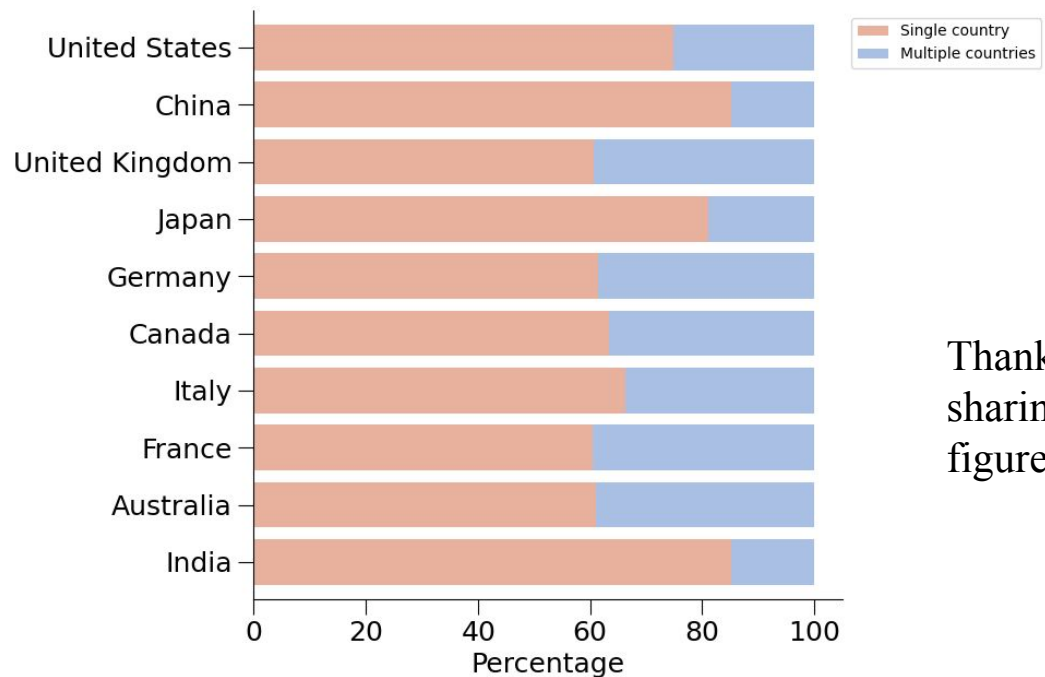
Articles	Journals	Authors	Affiliations	Countries
48518476	50797	38819457	56465	218



Thanks to Huaxia Zhou for sharing these statistics and figure on OpenAlex.

a.**b.**

OpenAlex (top),
Scopus (bottom)

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